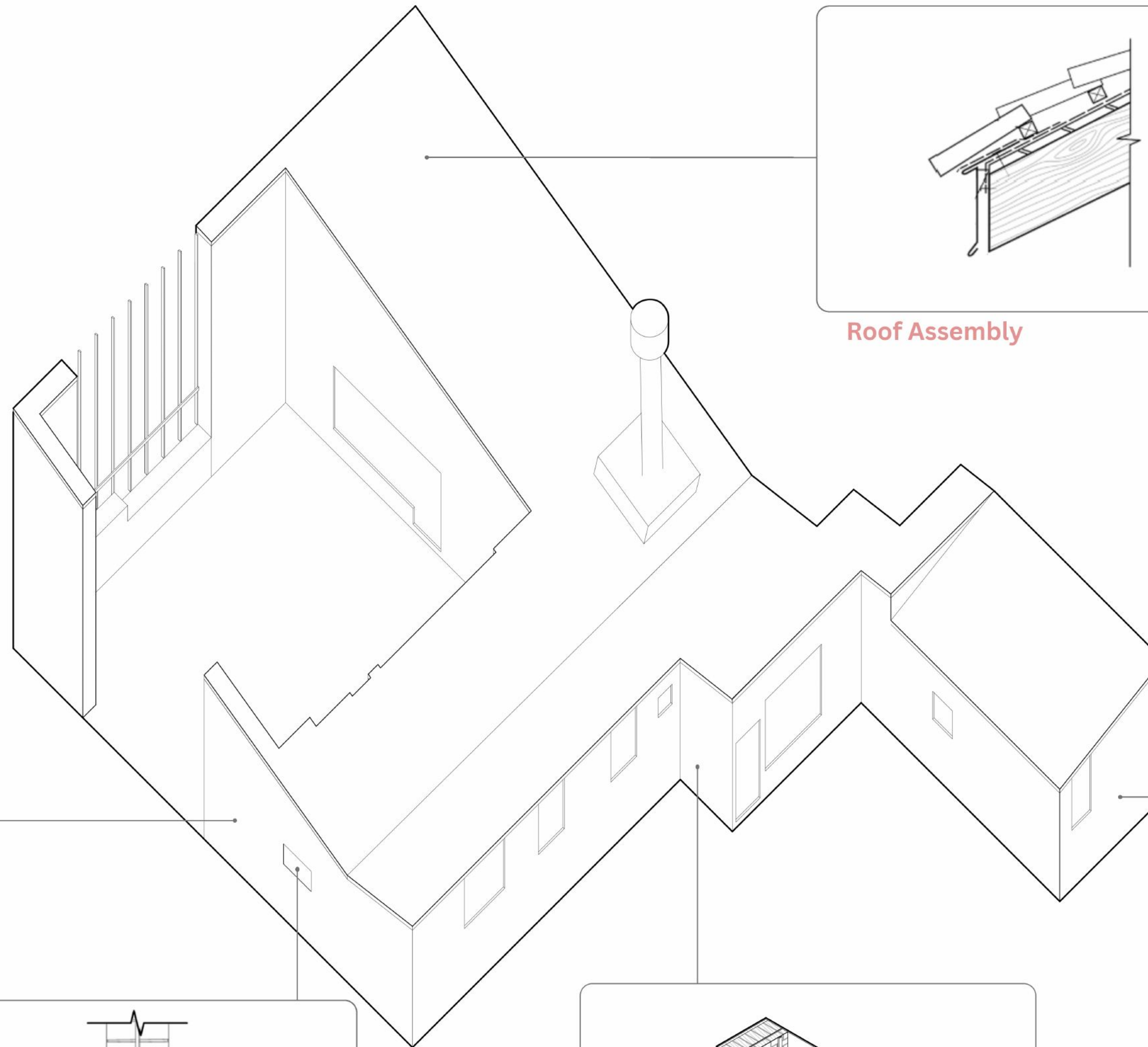
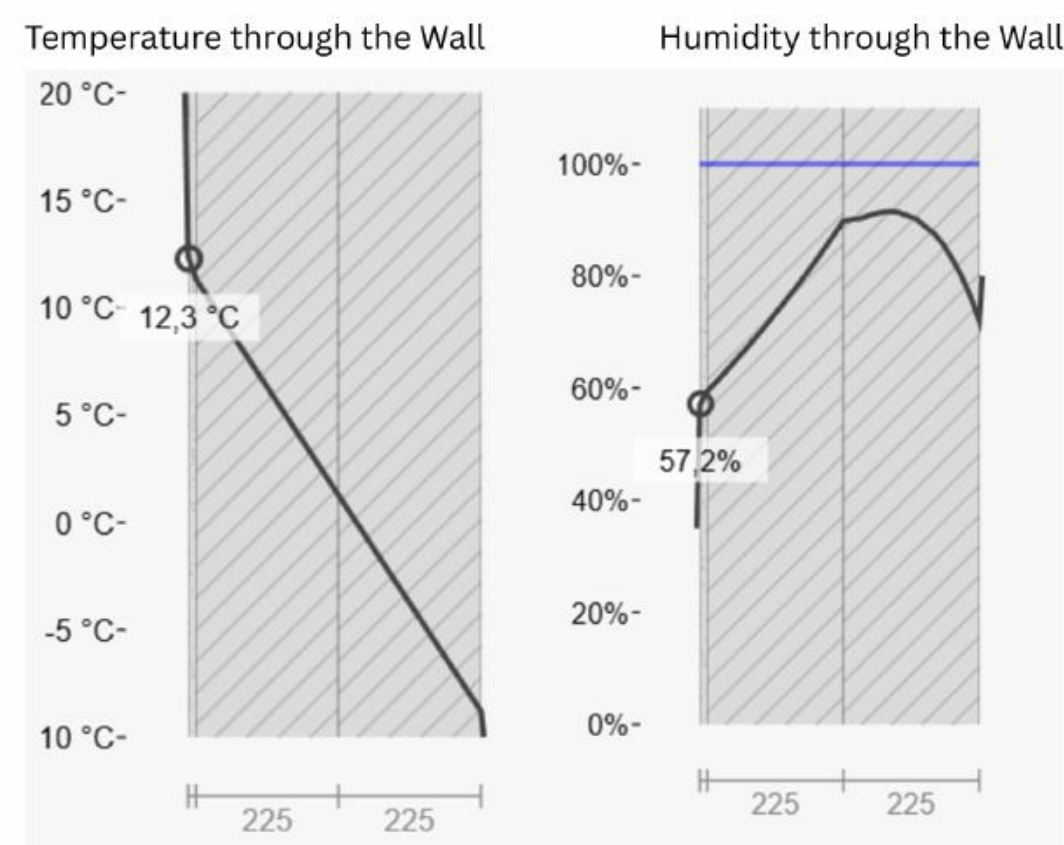
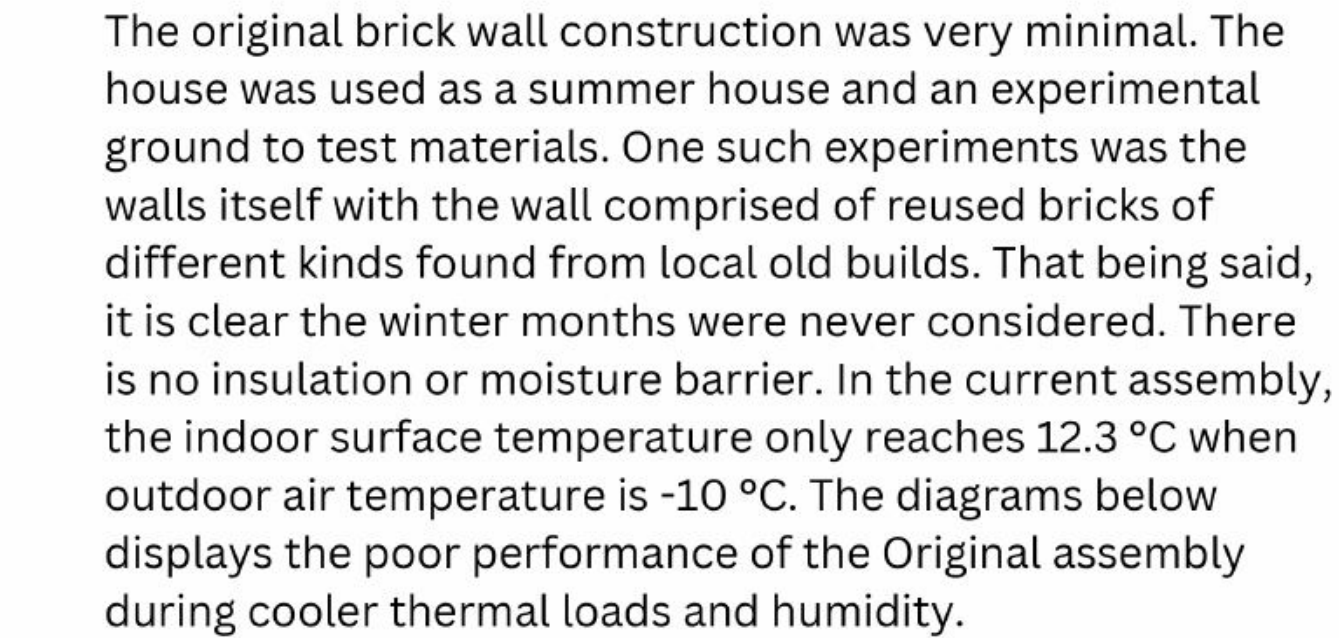


The envelope of the Experimental House is made of a collection of distinct wall, roof, and window assemblies. When viewed together, these components demonstrate the experimental and iterative nature of Aalto Alvars design. The mapping shown highlights the original envelope's varied Assemblies and locations on the building.



Technical drawing of a wall section. The left side shows a vertical scale with horizontal lines indicating height. The right side shows a brickwork pattern, likely representing a half-brick wall, with a vertical line indicating the edge of the wall.



The diagram shows a cross-section of a wall assembly with the following layers from outside to inside:

- Outside
- 8: solid brick 1600 kg/m³, DIN 105 (225mm)
- 7: Rockwool, stone wool (200mm)
- 6: 3-layer panel (10mm)
- 5: Spruce (180x100mm²)
- 4: Rockwool, stone wool (180mm)
- 3: 3-layer panel (10mm)
- 2: bitumen coating
- 1: solid brick 1600 kg/m³, DIN 105 (225mm)
- Inside

Thermal resistance values (R [m²K/W]) are listed on the right:

- 0.04
- 0.331
- 5
- 0.077
- 1.385
- 4.5
- 0.130
- 0.170
- 0.331
- 0.036
- 0.130

Dimensions: 865,5 (height), 100 (width), 600 (width).

Source: ubakus.de

Total Thermal Resistance of component + Surface Air Resistance = 9.639
Total U-Value [W/m²K] = 0.104

With the gap between the brick, the humidity is given the opportunity to reach 100% which leads to condensation between the brick and insulation. Due to this occurrence, adding in a moisture barrier was necessary. In the thermal renovations, bitumen coating was used for the purpose of the moisture barrier.

The graph shows the surface temperature during the day. The y-axis represents temperature in °C, ranging from 16 to 34. The x-axis represents the time of day, ranging from 12 to 12. The 'Outside' temperature (red line) fluctuates between approximately 16.5°C and 32.5°C. The 'Inside' temperature (blue line) is constant at 25°C.

Moving the Experimental House to Portland, Oregon, USA, the wall needs to be renovated to meet ICC code. To keep the original aesthetic and staying true to the houses' character defining elements, insulation was placed in-between the two sections of brick. This was supported with studs and wood panels. The R-value increases from 0.867 to 9.639 [m²K/W] after the changes. This far exceeds the ICC codes.

Layer	Thickness	Thermal Resistance R [m²K/W]
Outside		0.04
solid brick 1600 kg/m³, DIN 105	225mm	0.331
AGEPAN Flex	180mm	4.737
3-layer panel	10mm	0.077
Spruce (160x100mm²)		1.231
AGEPAN Flex	160mm	4.211
3-layer panel	10mm	0.077
Pfleiderer PremiumBoard Vapor P5	13mm	0.108
solid brick 1600 kg/m³, DIN 105	225mm	0.331
Gypsum plaster	12,5mm	0.036
Inside		0.130

Total Thermal Resistance of component + Surface Air Resistance = 9.220
Total U-Value [W/m²K] = 0.108

The gap between the brick provides the opportunity for humidity to reach 100% showcasing the same issue when considering the thermal analysis. A moisture barrier was needed. Premium Board Vapor P5 has a water vapor diffusion resistance factor of 600. Unfortunately it is not as effective and condensation does occur between the insulation and brick

When approaching the renovation design with a carbon conscious perspective, the insulation was changed to AGE-PAN Flex with a heat capacity of 2100 [J/(kg*K)]. The Insulation is supported by the wood studs and panels. The R-value increases from 0.867 to 9.220 [m²K/W] after the changes. This far exceeds the ICC codes.

In the original construction bricks were reused from local sites for a unique visual and reducing the green house gases that will be produced in the manufacturing of new materials. These bricks will still be used in the renovations.

The overall GHG potential is -11 kg CO₂ Äqv./m². The spruce used for the studs has a reduction potential of -18 kg and the rockwool has a GHG potential of 18 kg effectively reaching a net zero. The leftover -11 kg is from the wood panels and bitumen coating.

The non renewable primary energy used is 125 kWh/m² with the rockwool contributing 65%.

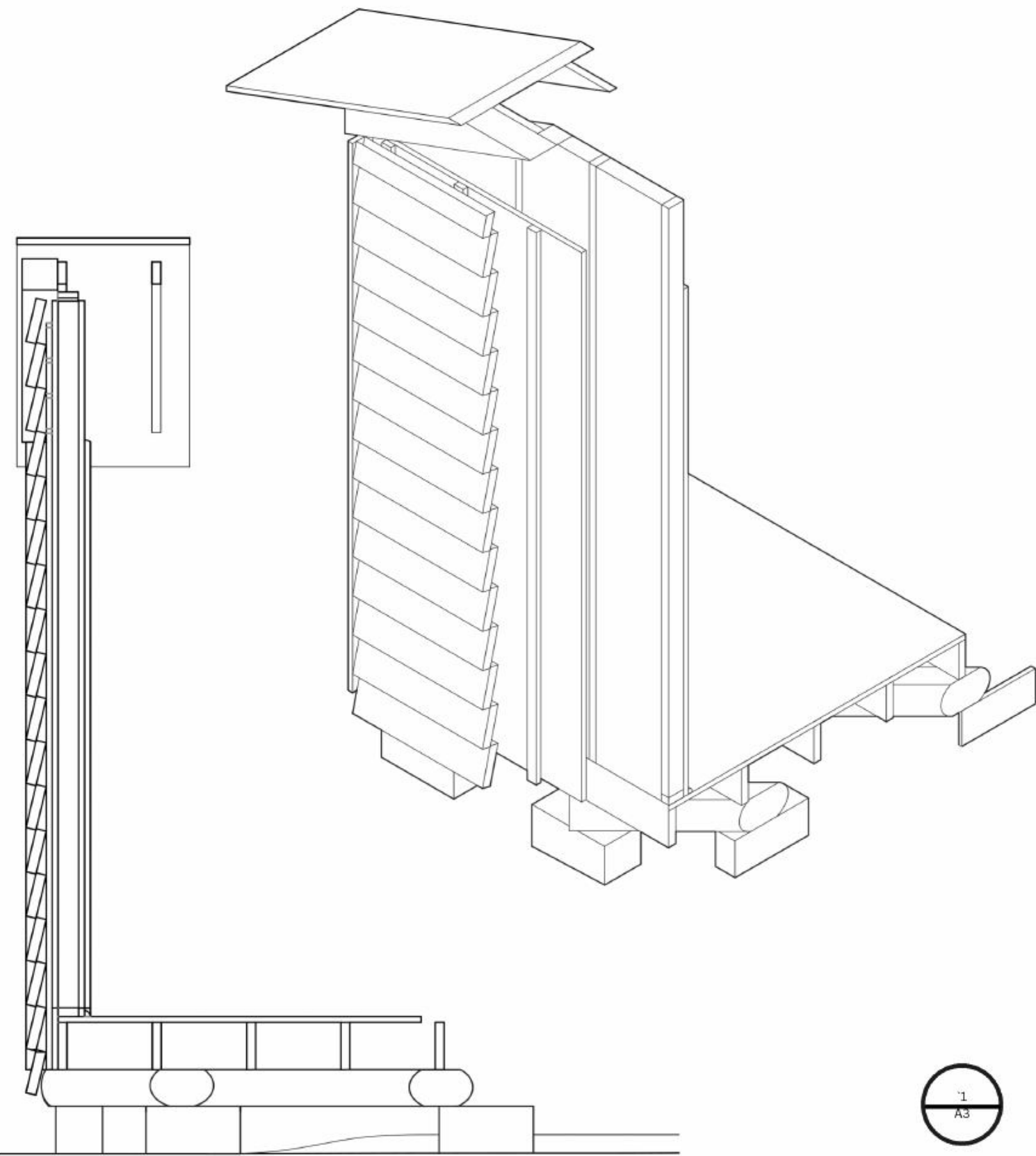
As stated in the thermal performance assembly analysis, the bricks used in this construction were reused in the original construction and will not be replaced in the renovations.

The overall GHG potential is -55 kg CO₂ Äqv./m². All materials used in this construction have a negative GHG potential.

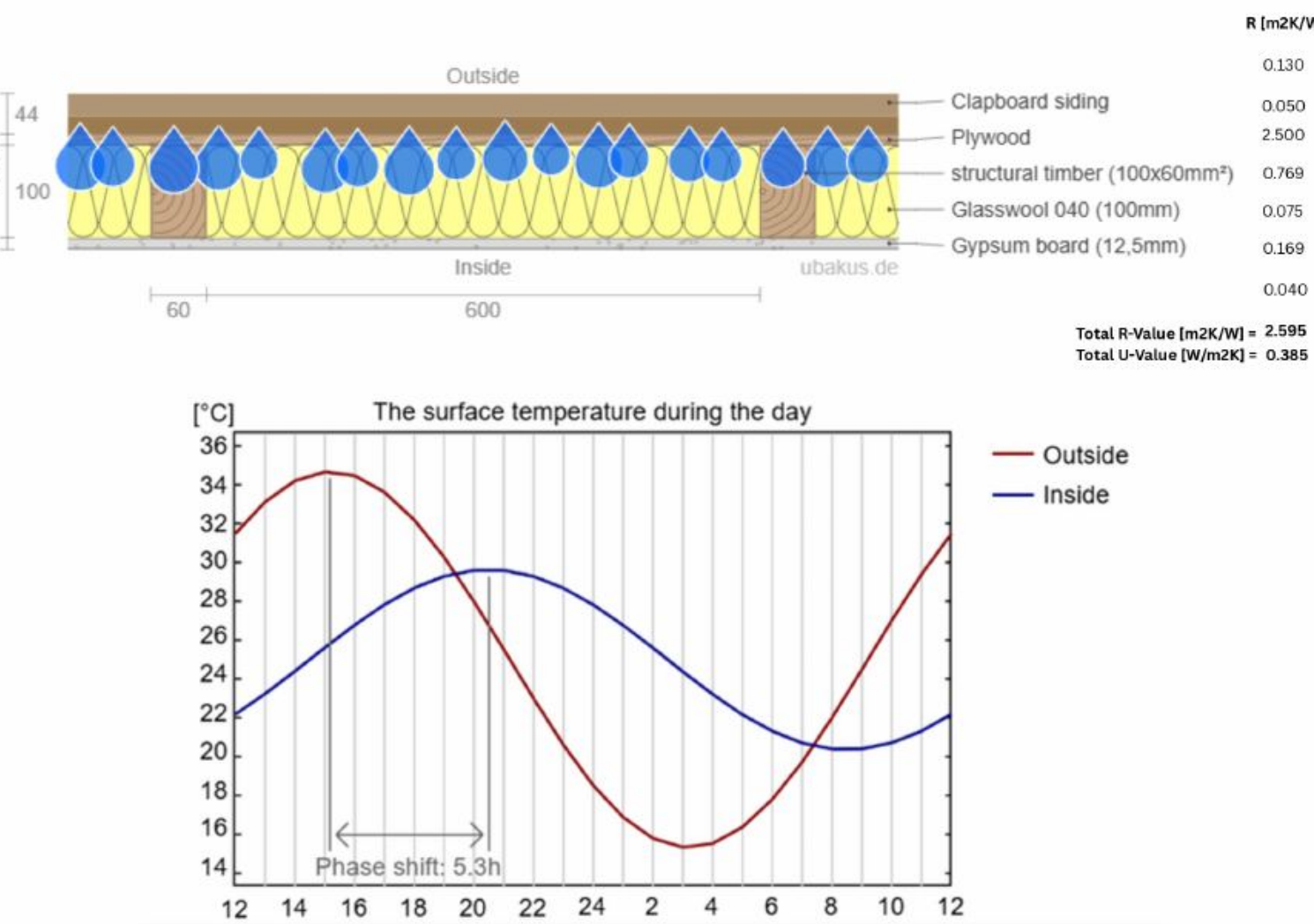
The non renewable primary energy used is 68 kWh/m² with the AGEPAN Flex contributing 61%.

CARLETON UNIVERSITY AZRIELI SCHOOL OF ARCHITECTURE AND URBANISM	EXPERIMENTAL HOUSE MELALAMMETIE, 40900, FINLAND	ENVELOPE RE(DESIGN) AND IMPACT ASSESSMENT	SHEET NUMBER:	TEAM MEMBER: Manreet Atwal	REVISION SCHEDULE:	Manreet Atwal	11-18-2025	Added Brick Wall Analysis, Diagrams, Drawing, and Graphs
	DATE OF CONSTRUCTION 1952-21954		A2			Manreet Atwal	11-24-2025	Updated Details, and 3D Drawings
						Manreet Atwal	11-25-2025	Added in depth moisture, thermal, and LCA

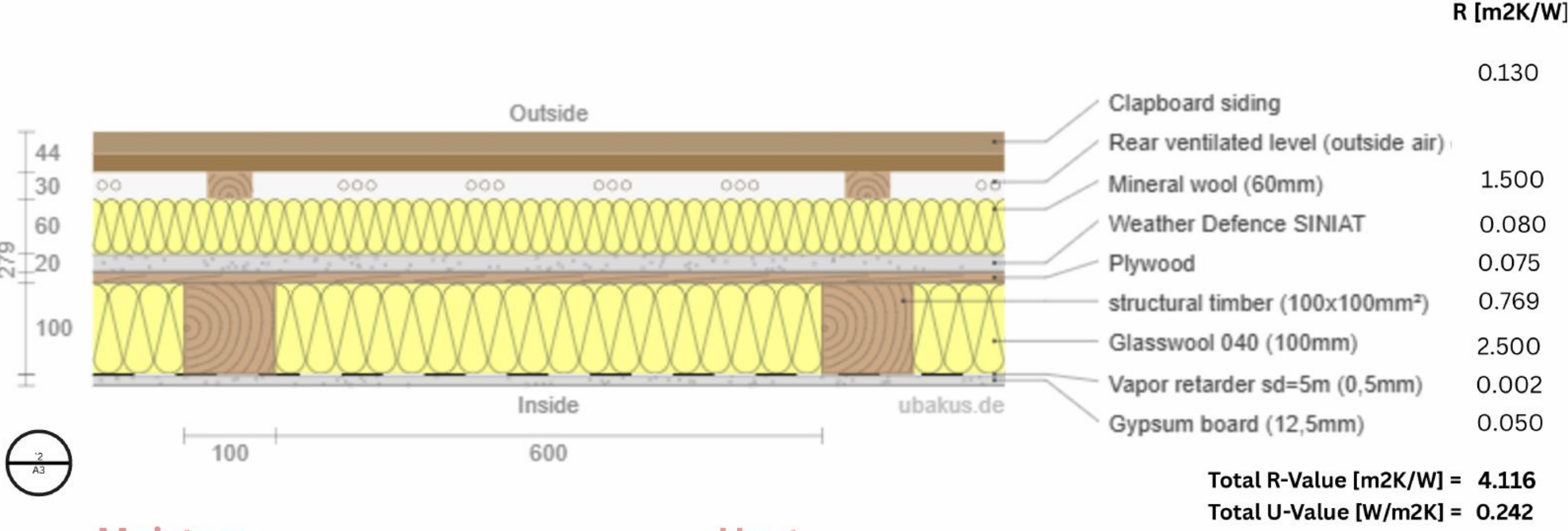
ORIGINAL VINYL WALL ASSEMBLY



The building's light wood-frame wall does not perform well in Portland's warmer and sunnier climate. With a U-value of 0.385 W/m²·K and limited insulation, it offers weak thermal resistance and risks winter condensation. Although Portland has milder temperatures and consistent winds, making it usable year-round, the original wall assembly does not manage heat loss or moisture effectively. While Portland reduces cold stress and provides more natural light, the existing wall still does not meet comfort and durability needs. This situation shows the importance of using a better-insulated, moisture-resistant wall design that suits the local climate.



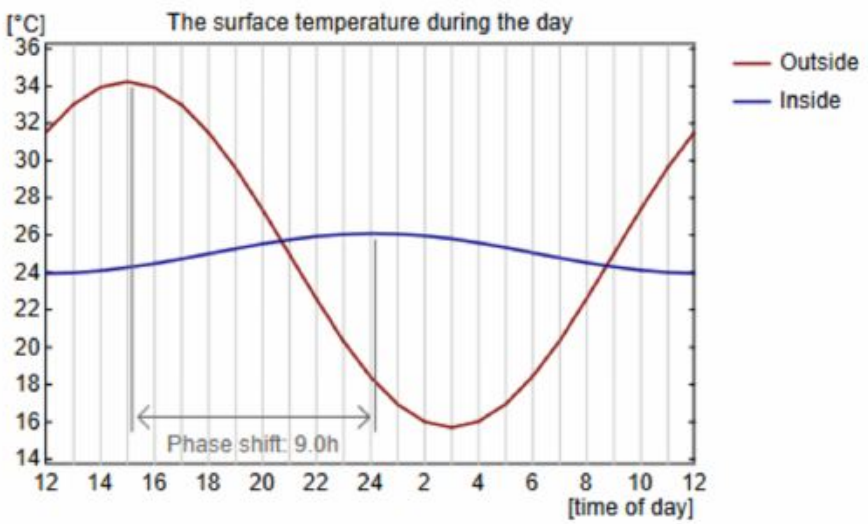
THERMAL PERFORMANCE VINYL WALL ASSEMBLY



Moisture

The original vinyl wall struggled with moisture control, offering little protection against Portland's humid winters and increasing the risk of condensation inside the assembly. The updated design adds a vapour retarder, a diffusion-open Weather Defence layer, and a rear-ventilated cavity to allow moisture to dry outwards efficiently. These features reduce the risk of mould, rot, and durability issues by better managing moisture in Portland's climate.

Heat



The new design creates a clearer delay between outdoor temperature peaks and the corresponding indoor response, indicating better thermal lag and damping. Heat moves more slowly through the assembly, resulting in a smoother and more stable indoor temperature curve.

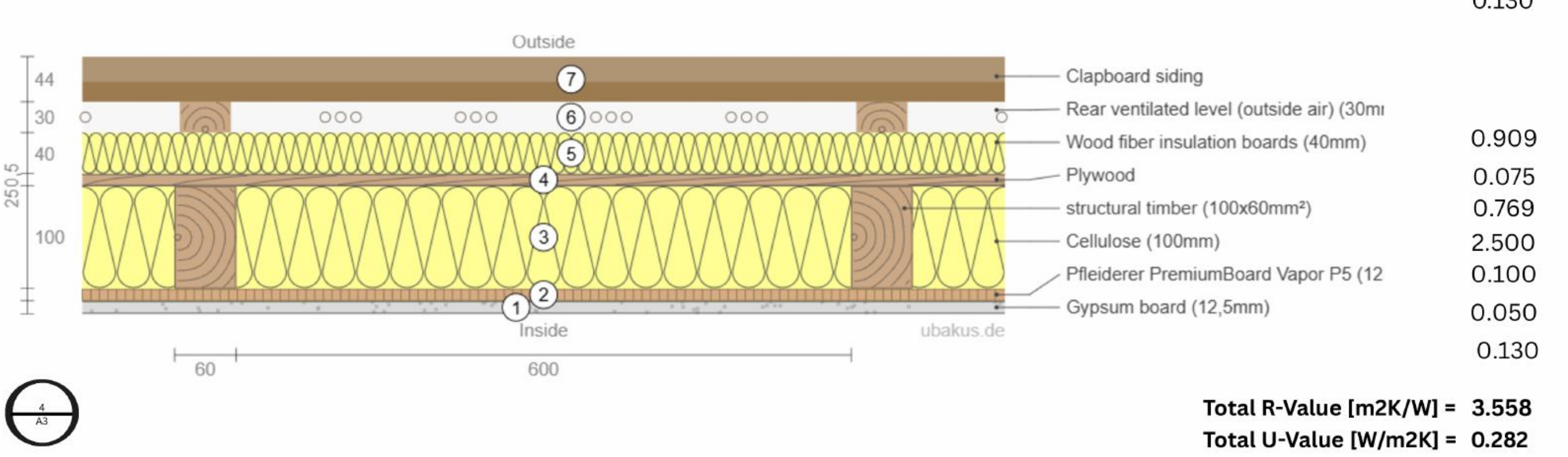
U-Value

The U-value of the original wall assembly reflects its limited insulation and poor thermal resistance. The wall now achieves a lower U-value of 0.243 W/(m²·K), corresponding to an overall thermal resistance of R = 4.116 m²·K/W. This reduction means far less heat escapes during colder periods and far less heat enters during Portland's warmer afternoons, providing a more stable and comfortable indoor environment year-round.

LCA

The original wall design had little insulation, a shorter lifespan, and required more heating energy over time. This new design lasts longer and uses less energy. It improves insulation, manages moisture better, and includes a ventilated cladding system that protects the materials from damage. Although mineral wool and glass wool have some carbon costs, the overall impact on the environment is still lower than the original design. This is because the new assembly uses less energy in the long run and extends the life of the materials, making it a more sustainable choice for Portland's climate.

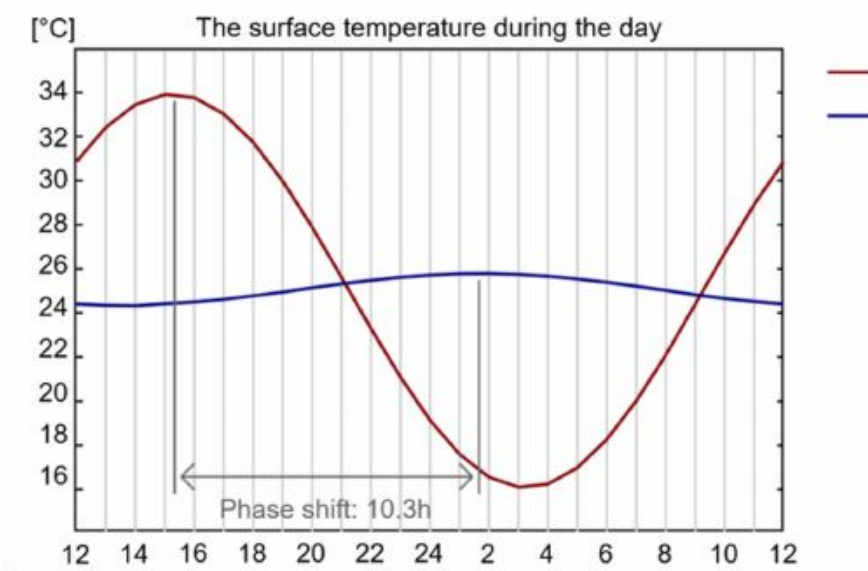
CARBON CONSCIOUS VINYL WALL ASSEMBLY



Moisture

This wall assembly effectively manages moisture with a vapour-smart membrane that limits winter vapour diffusion and allows inward drying in summer. The cellulose insulation holds and releases moisture, while the vapor-open wood fiber insulation protects the plywood and enables outside drying. A rear ventilated cavity provides airflow to remove moisture, preventing mold, decay, and long-term damage.

Heat



The graph indicates strong thermal damping with a delay of roughly 10.3 hours, as heat takes time to travel through the wall. The insulation, particularly cellulose and wood fiber, effectively slows heat transfer, stabilizing the indoor environment, enhancing comfort, and reducing heating and cooling needs.

U-Value

The wall's U-value and R -value indicate strong heat resistance that outperforms standard timber-frame walls. Continuous wood fiber insulation minimizes thermal bridging, while cellulose-filled cavities offer excellent insulation using renewable materials. Together, these elements enhance thermal efficiency and promote energy-efficient buildings.

LCA

The wall assembly is good for the environment because it uses natural and renewable materials, like cellulose, wood fiber insulation, and timber. These materials store carbon and use less energy to produce than mineral or petrochemical products. Although plywood has some impact due to its adhesives, it is still better than synthetic sheathing. The vinyl siding is the least eco-friendly part because it comes from fossil fuels and is hard to recycle. Overall, this wall has a positive carbon profile and supports sustainable building practices.

CARLETON UNIVERSITY
AZRIELI SCHOOL OF
ARCHITECTURE AND
URBANISM

EXPERIMENTAL HOUSE
MELALAMMETIE, 40900, FINLAND

DATE OF CONSTRUCTION
1952-21954

ENVELOPE RE(DESIGN)
AND IMPACT
ASSESSMENT

SHEET NUMBER:

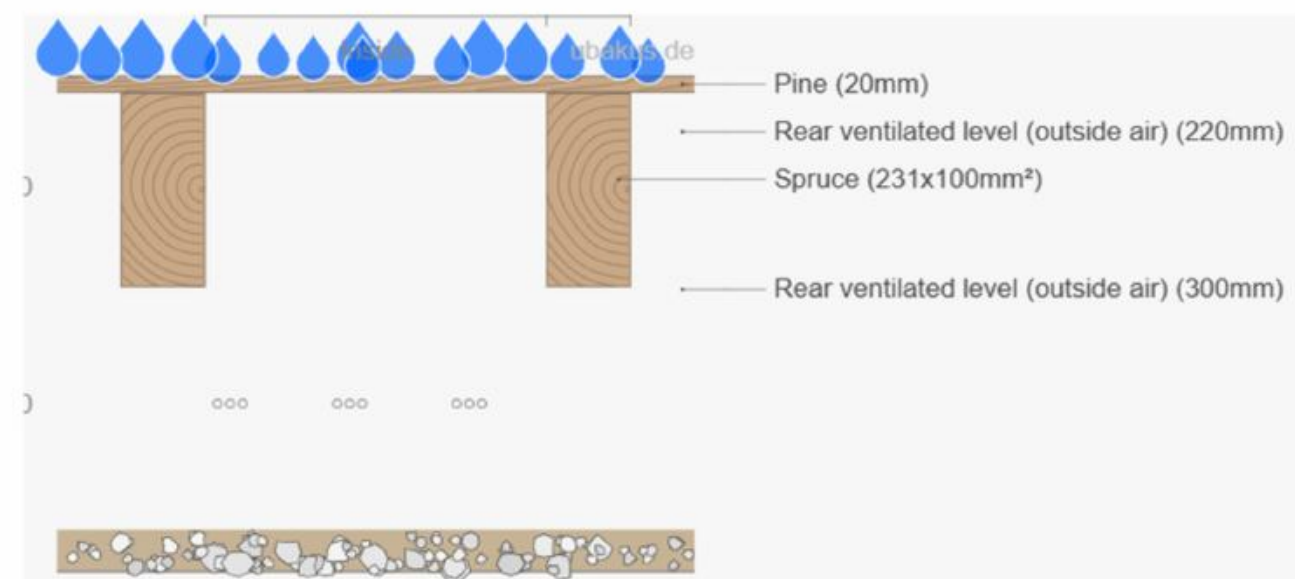
A3

TEAM MEMBER:

Sofia Canas Morales

REVISION SCHEDULE:

11/12/2025	Sofia Canas Morales	Created a Ubakus file for the original vinyl wall assembly
11/18/2025	Sofia Canas Morales	Added section and perspective drawing to display original wall detail
11/18/2025	Sofia Canas Morales	Added Ubakus file of thermal performance wall assembly
11/18/2025	Sofia Canas Morales	Wrote analysis on the original and thermal performance wall assembly
11/19/2025	Sofia Canas Morales	Re-evaluated thermal and carbon siding wall assembly following pin up
11/23/2025	Sofia Canas Morales	Modified drawings and general sheet layout after pin ups and TA feedback
11/25/2025	Sofia Canas Morales	Performed analysis regarding heat and moisture for different variations of the wall assembly



The original floor system of the home was very limited, the structure was just enough to support the demand from above with no additional insulation. Since this home was only used the summer, heat losses from the interior to the ground was not a huge concern during colder months. As shown in the graph below, the temperature of the interior of the home is directly related to the outside temperature. This is due to a low USI-value of 0.506. The original site was very rocky and well ventilated due to the foundation walls and therefore no vapor barrier was required.

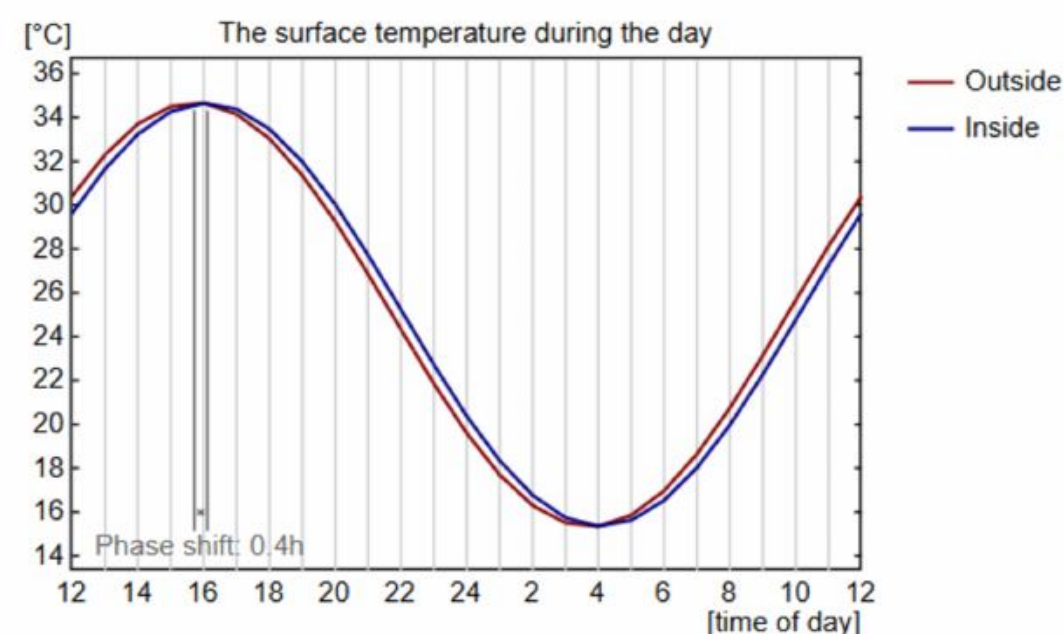


Diagram illustrating the cross-section of a roof assembly, showing the following layers from top to bottom:

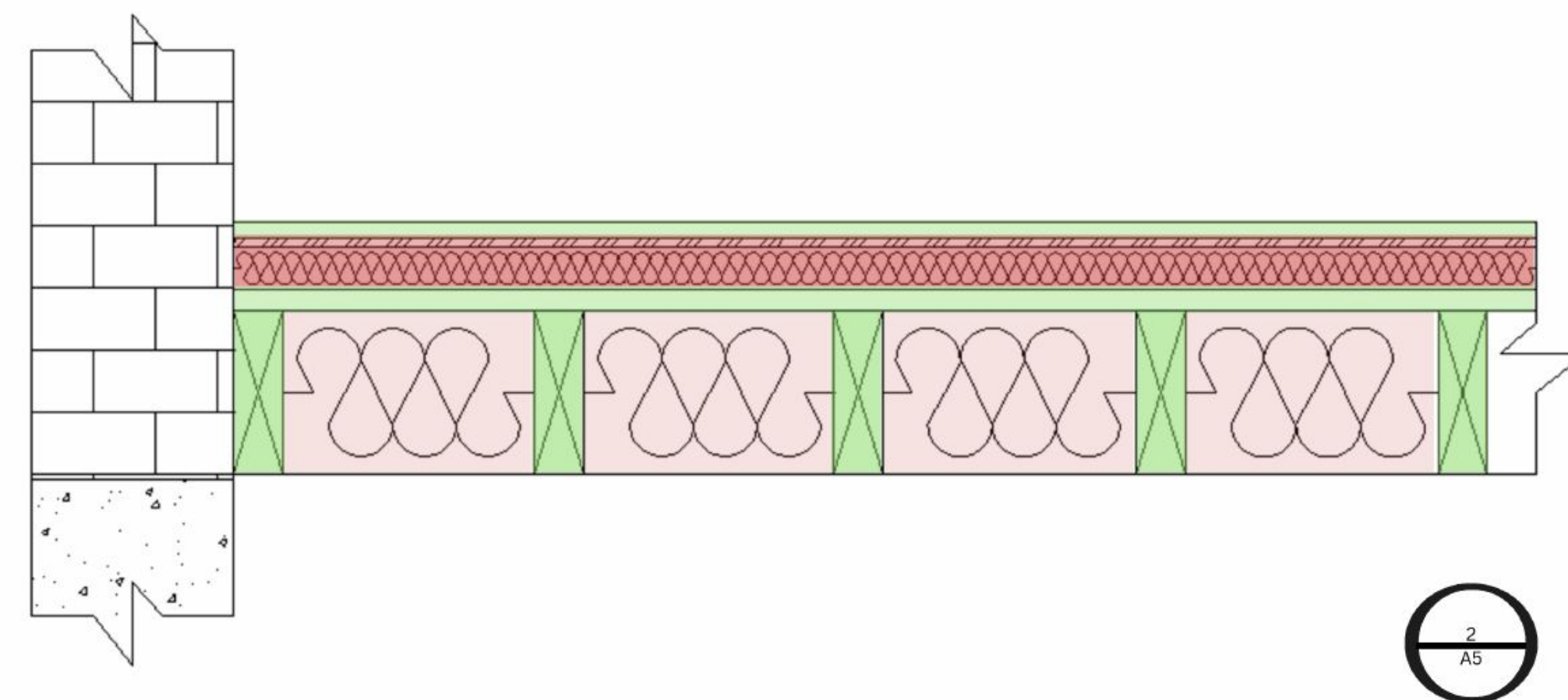
- Pine (20mm) (wood planks)
- Polyethylene, high density (10mm)
- Vacuum insulation (40mm)
- Pine (20mm)
- Rockwool, stone wool (200mm)
- Pine (203x50mm²)
- Rear ventilated level (outside air) (300mm)

The diagram also shows three small circles (○○○) representing ventilation points and a cross-section of the rear ventilated level (outside air) at the bottom.

The new soil conditions feature softer soil compared to Finland and therefore a vapour barrier must be added to protect moisture getting into the home. The vapour barrier was added below the floor boards but above all layers of insulation to prevent condensation and mold growth.

The graph, titled "The surface temperature during the day", plots temperature in degrees Celsius [°C] on the y-axis (ranging from 14 to 36) against time of day on the x-axis (ranging from 12 to 12). Two data series are shown: "Outside" (red line) and "Inside" (blue line). The "Outside" temperature follows a sinusoidal wave, peaking at approximately 35°C around 16:00 and reaching a minimum of about 17°C around 04:00. The "Inside" temperature is a constant horizontal line at 25°C. A horizontal double-headed arrow indicates the "Phase shift: 14.7h" between the peak of the outside temperature and the constant inside temperature.

With proper insulation of the floor, the inside conditions do not vary leading to improved occupant comfort and lowered heating loads. The continuous layer of insulation limits the amount of thermal bridging that can occur.



The thermal performance of the floor assembly exceeds code requirement for marine zone according to IECC guideline. By adding 200mm of RockWool Stone insulation and 40mm of vacuum panels, the USI-value decreased to 0.095. The RSI-value for this layout is 10.52, The floor system will not be able to handle the Portland Winter climate .

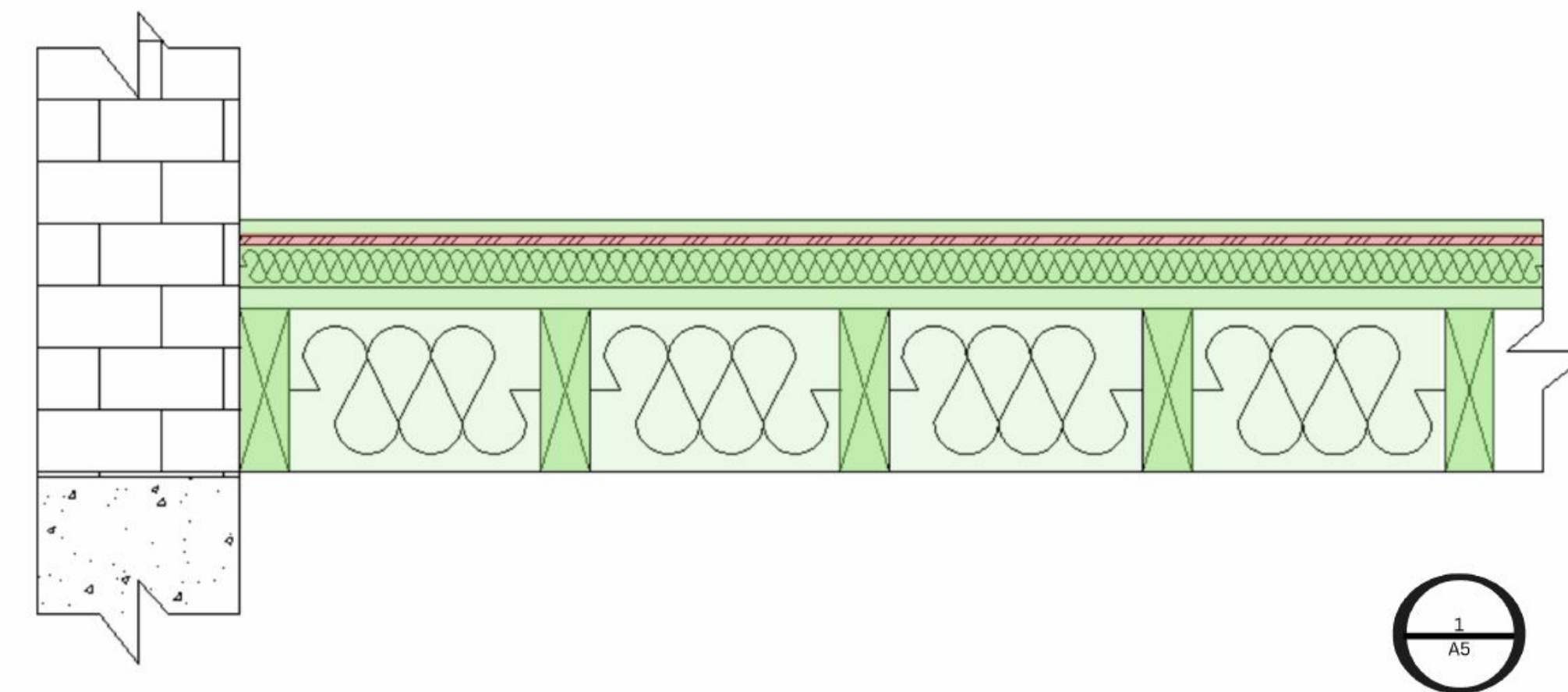
The colours on the diagram represent the associated global warming potential of the production in kg CO₂ eqv/m². The main contributor in this system are the vacuum panels that are made from steel. The stone wool has very low kg CO₂, while the wood materials are carbons sequesters highlighted in green.

Diagram illustrating the cross-section of a floor assembly, showing the following layers and components:

- Pine (20mm) (wood planks)
- Polyethylene, high density (10mm)
- STEICOfloor (40mm)
- Pine (20mm)
- AGEPAN Flex (200mm)
- Pine (203x50mm²)
- Rear ventilated level (outside air) (300mm)

The same polyethylene vapor barrier was kept in the carbon conscious assembly as this is a product that must perform well. Without an adequate barrier, moisture will damage with wood and insulation leading to increased carbon emissions due to replacement.

The phase shift of the carbon conscious floor assembly is less than that of the thermal performance. There is a slight variation of the indoor temperature very early in the morning due to inadequate insolation for these conditions.

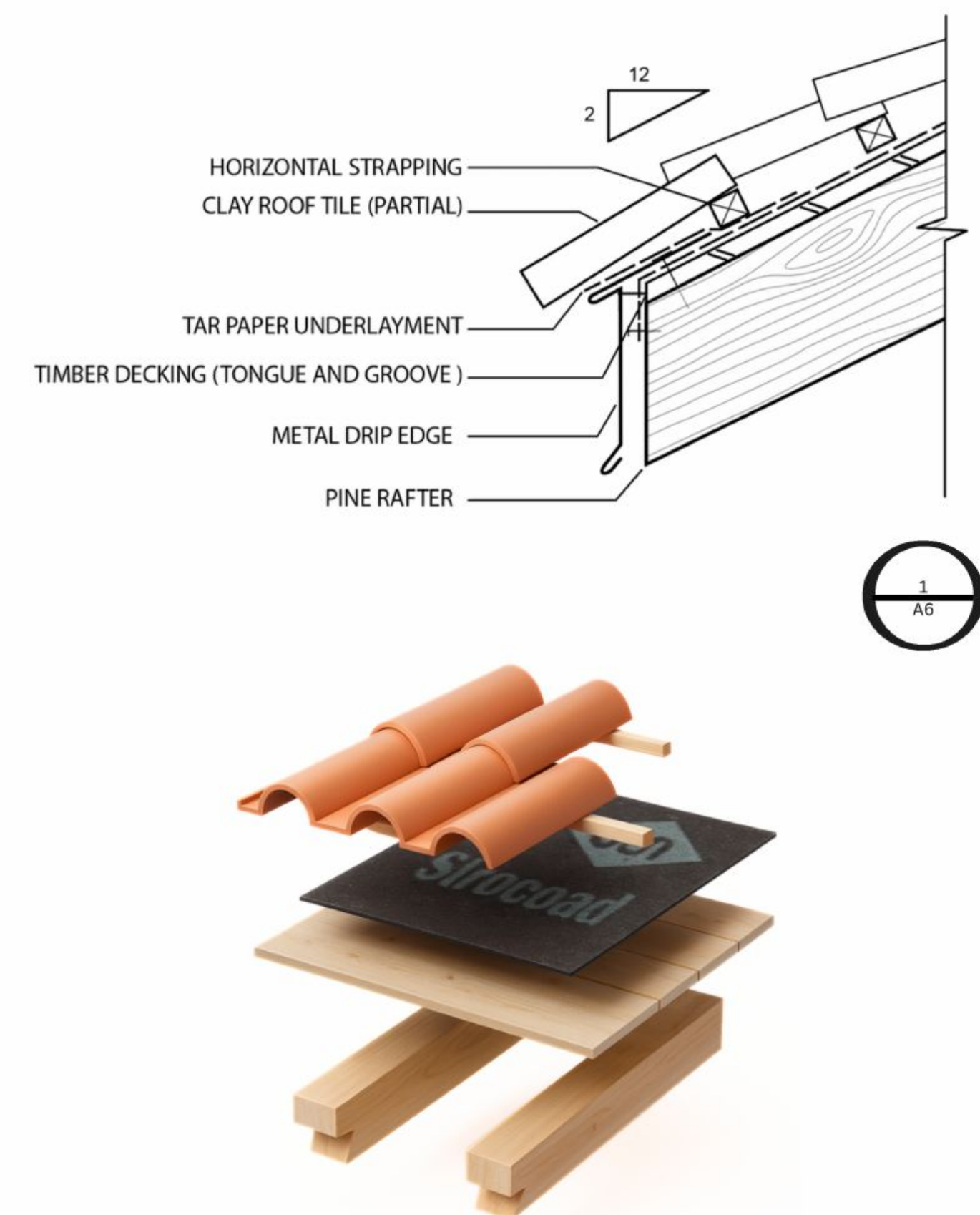


The insulation between the floor joist is AGEPAN, a wood based form of batt insulation manufactured in Europe. Even though this product has less CO₂ eqv/m², we must consider the transportation emissions included to get this product to site. STEICOfloor is manufactured in U.S.A. contributing less CO₂ emissions in the transportation stage.

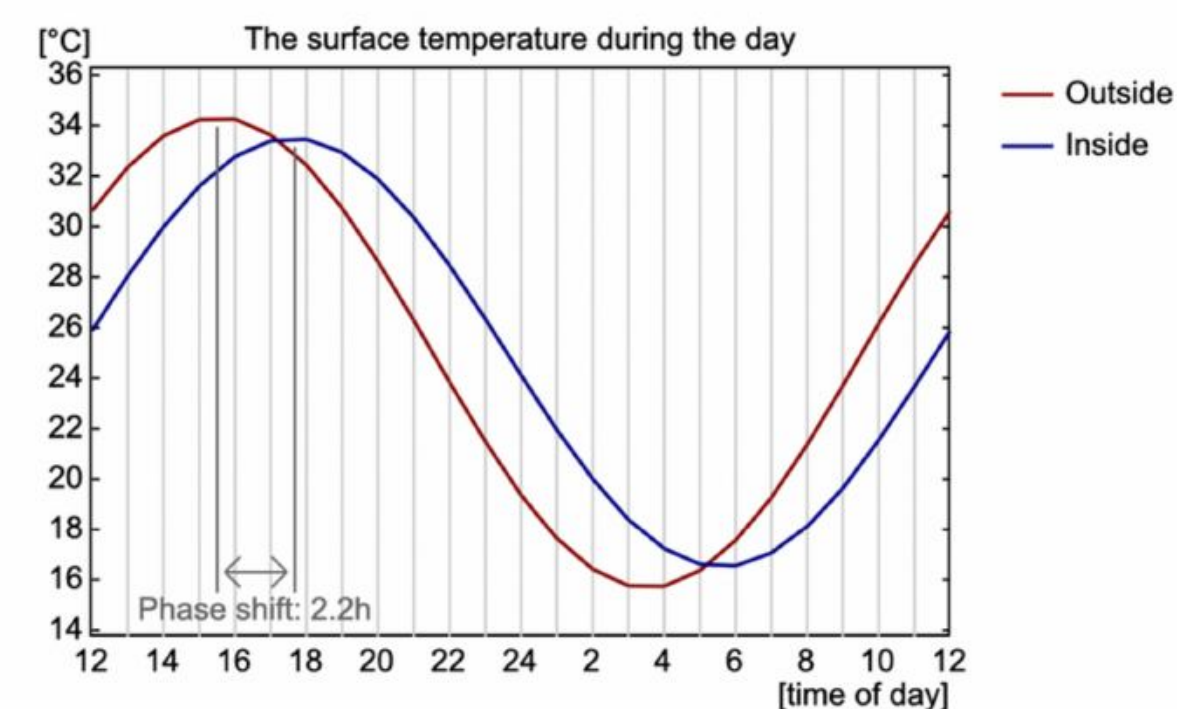
The thermal performance of the floor assembly does meet ICC code. The USI-value is 0.0164. AGEPAN was added instead of Rockwool as it is made from wood fibers and provides a high R-value. The steel vacuum panels were replaced with STEICOfloor, another wood based insulation product that omits glue or and laminate productions from production.

CARLETON UNIVERSITY AZRIELI SCHOOL OF ARCHITECTURE AND URBANISM	EXPERIMENTAL HOUSE MELALAMMETIE, 40900, FINLAND	ENVELOPE RE(DESIGN) AND IMPACT ASSESSMENT	SHEET NUMBER:	TEAM MEMBER: Kaitlin Gibson	REVISION SCHEDULE:	11/14/2025	Kaitlin Gibson	Began analysis of floor structure using Ubakus tool
	DATE OF CONSTRUCTION 1952-21954		A5			11/15/2025	Kaitlin Gibson	Created 2D and 3D drawings to match existing and new floor assemblies
						11/19/2025	Kaitlin Gibson	Re-evaluated thermal and carbon floor plan following pin up
						11/19/2025	Kaitlin Gibson	Performed analysis regarding heat and moisture for different variations of the floor assembly

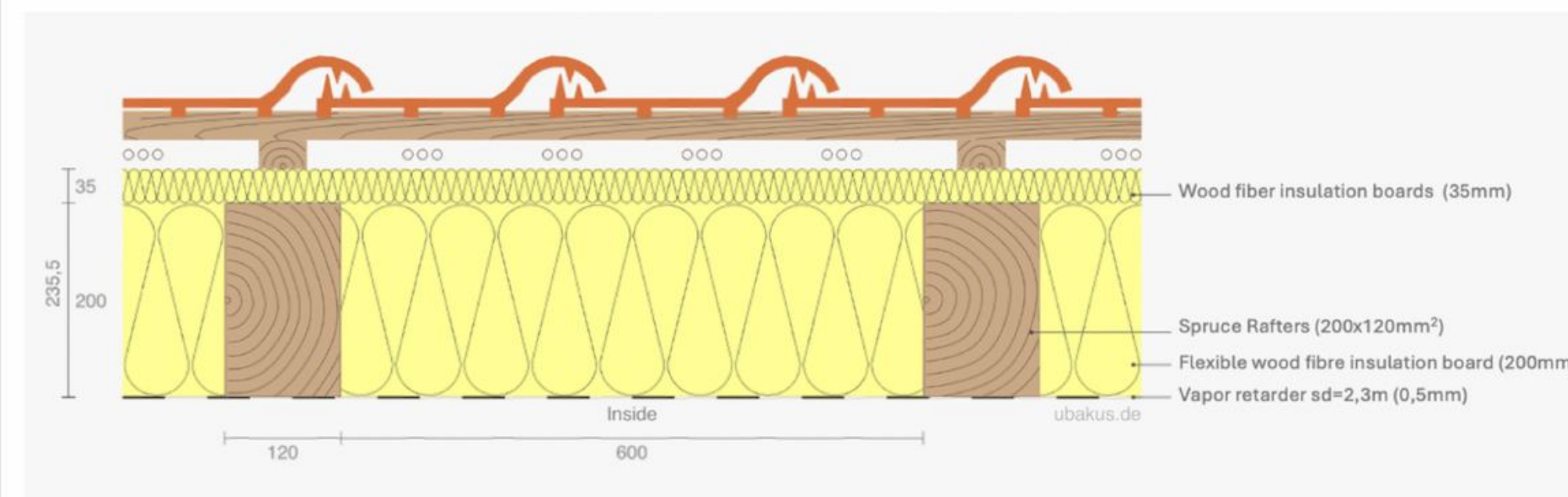
ORIGINAL ROOF ASSEMBLY



The construction of the original roof on the Alvar Aalto house includes spruce rafters supporting a tongue and groove deck, covered with a bituminous underlayment and scattered clay tiles. Architecturally, this minimal assembly aligns with Aalto's experimental, landscape-driven approach. primary environmental impacts occur during manufacturing and raw material sourcing of the clay. its performance weaknesses become clear when analyzed thermally. The absence of insulation and moisture performance is limited due to the bituminous layer providing weak vapour control and the scattered shingles.



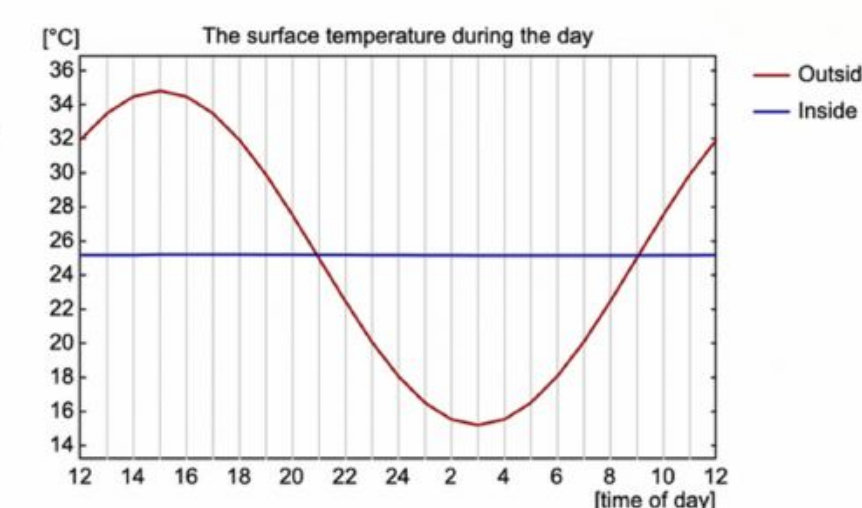
THERMAL PERFORMANCE ROOF ASSEMBLY



MOISTURE

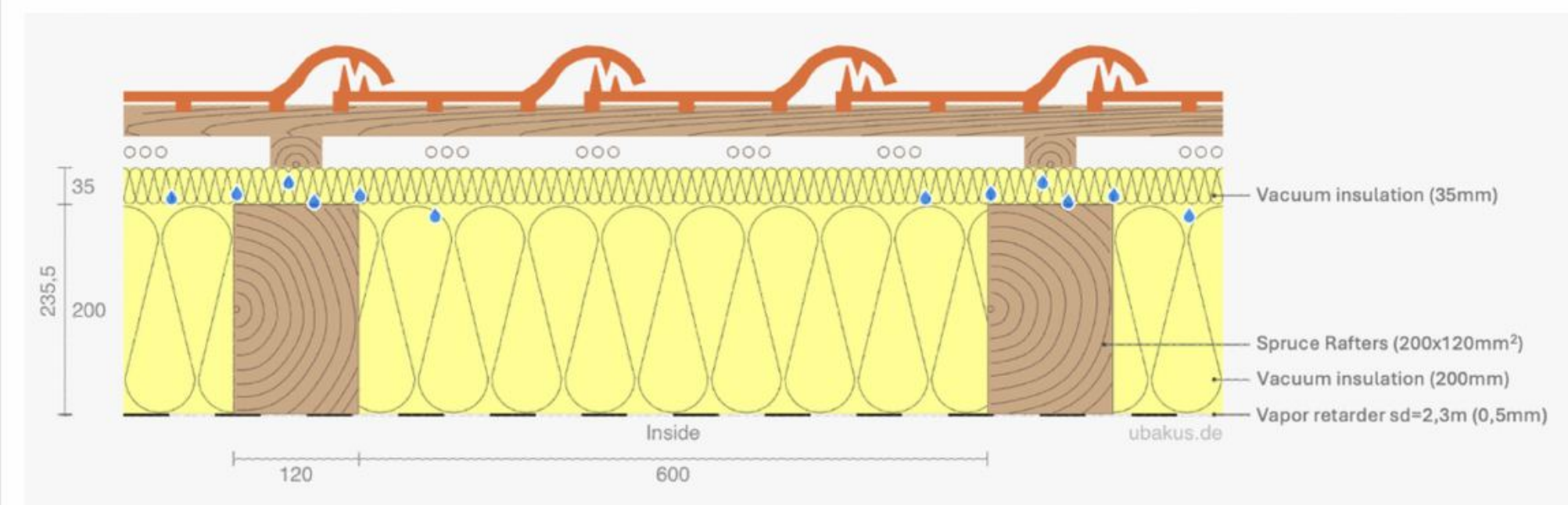
The high-performance roof remains condensate-free because the vacuum insulation layers are nearly vapour-impermeable, giving the assembly an extremely high sd-value that blocks moisture from entering. This creates a very safe winter profile but no drying reserve, meaning the assembly depends entirely on the interior vapour retarder staying intact. If that layer is compromised, moisture cannot diffuse outward and may become trapped.

HEAT



The interior temperature line remains almost perfectly flat throughout the day, showing no meaningful response to exterior temperature swings. Heat takes so long to move through the assembly that daily outdoor fluctuations never reach the interior.

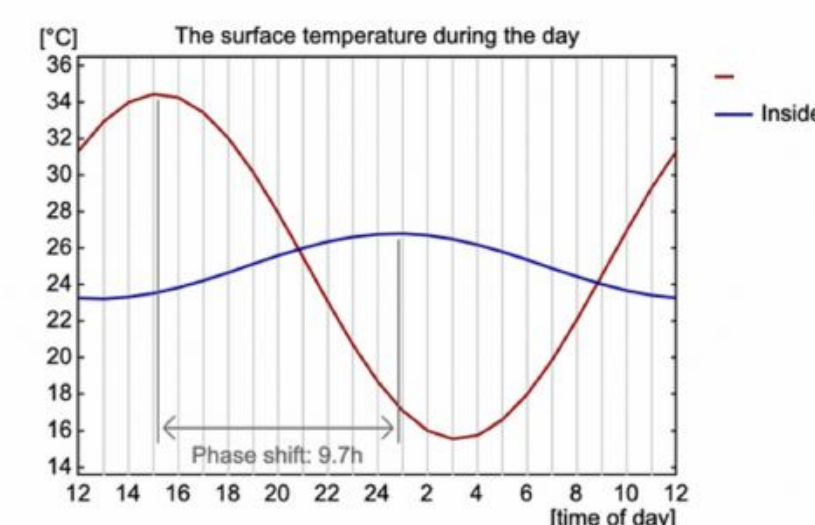
CARBON CONSCIOUS ROOF ASSEMBLY



MOISTURE

In the improved ecological roof, a vapour retarder on the warm side stops interior moisture from entering the thick wood-fibre insulation in winter, preventing condensation and mold. The remaining layers stay vapour-open, and the ventilated cavity beneath the clay tiles allows any moisture to dry outward in summer. This assembly stays condensate free year round while still using natural, carbon-storing materials.

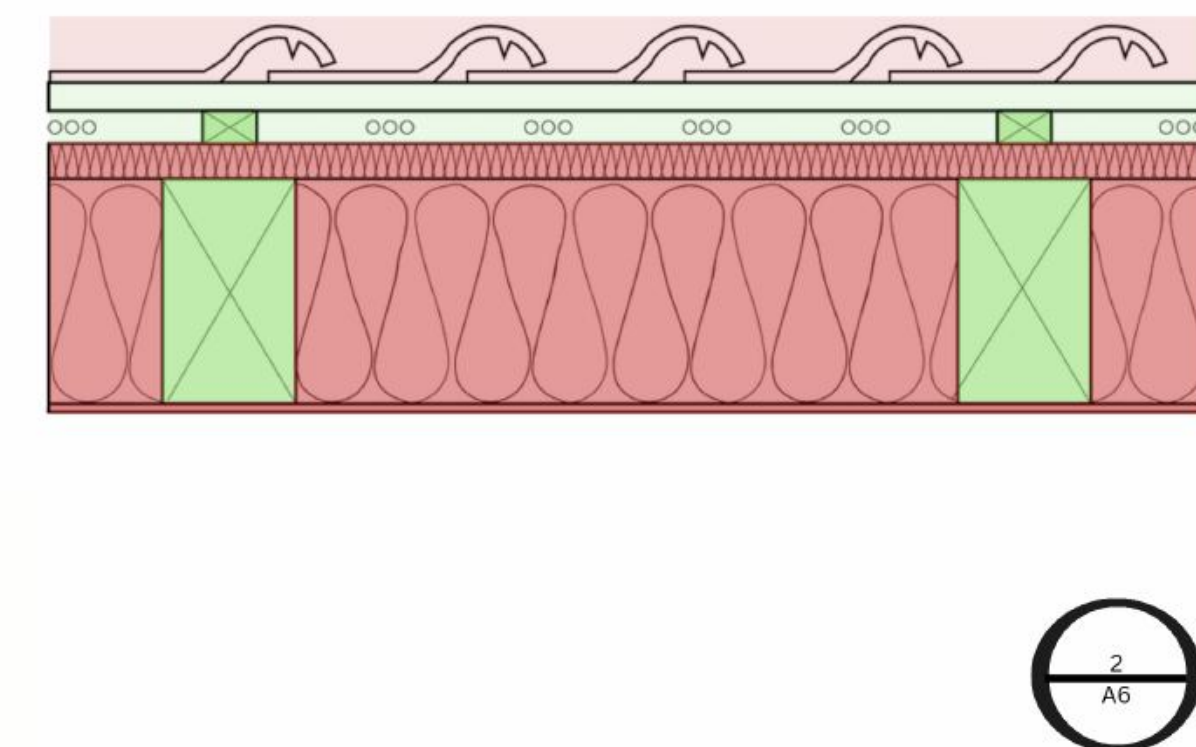
HEAT



The phase shift of the carbon-conscious roof assembly keeps indoor temperatures stable throughout the day. A small early-morning dip appears due to the lower R-value of wood-fibre insulation, yet the interior curve remains far more stable than the sinusoidal fluctuation of the original roof.

LCA

The high-performance roof has a very high embodied carbon footprint, as vacuum insulation panels require energy-intensive manufacturing and do not store biogenic carbon. Their environmental impact is fixed from production and increases over time as the vacuum layer slowly degrades. While the assembly performs extremely well thermally, it is the least ecological option due to its large cradle-to-gate emissions.

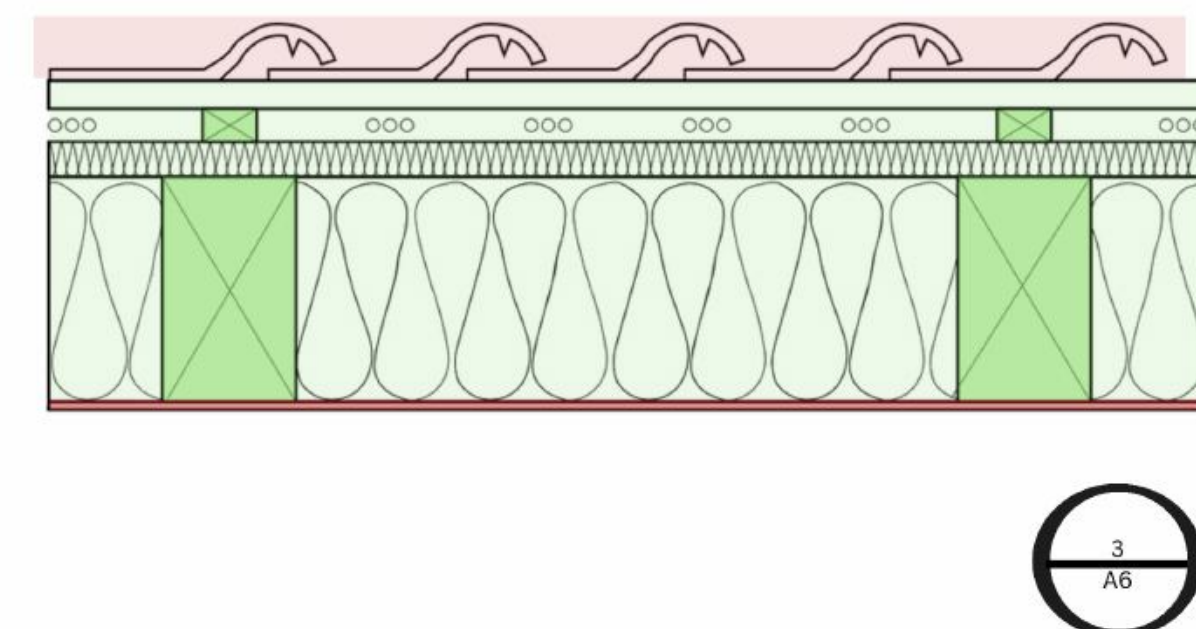


U-VALUE

The optimized thermal roof achieves a low U-value of $0.057 \text{ W/m}^2\text{K}$. This level of resistance is due to the vacuum insulation layers, which deliver high R-values. As a result, the assembly provides near-total thermal separation between interior and exterior conditions, keeping indoor temperatures stable with minimal energy input.

LCA

Wood-fibre insulation is an ecological choice for the roof assembly. Because it is made from sawmill by-products and fast-growing softwood, its production has a low embodied energy and stores carbon throughout its life. It naturally biodegrades at end-of-life, returning to the carbon cycle without generating toxic waste. It contributes negative CO₂ values, offsetting emissions from other components such as the clay tiles.

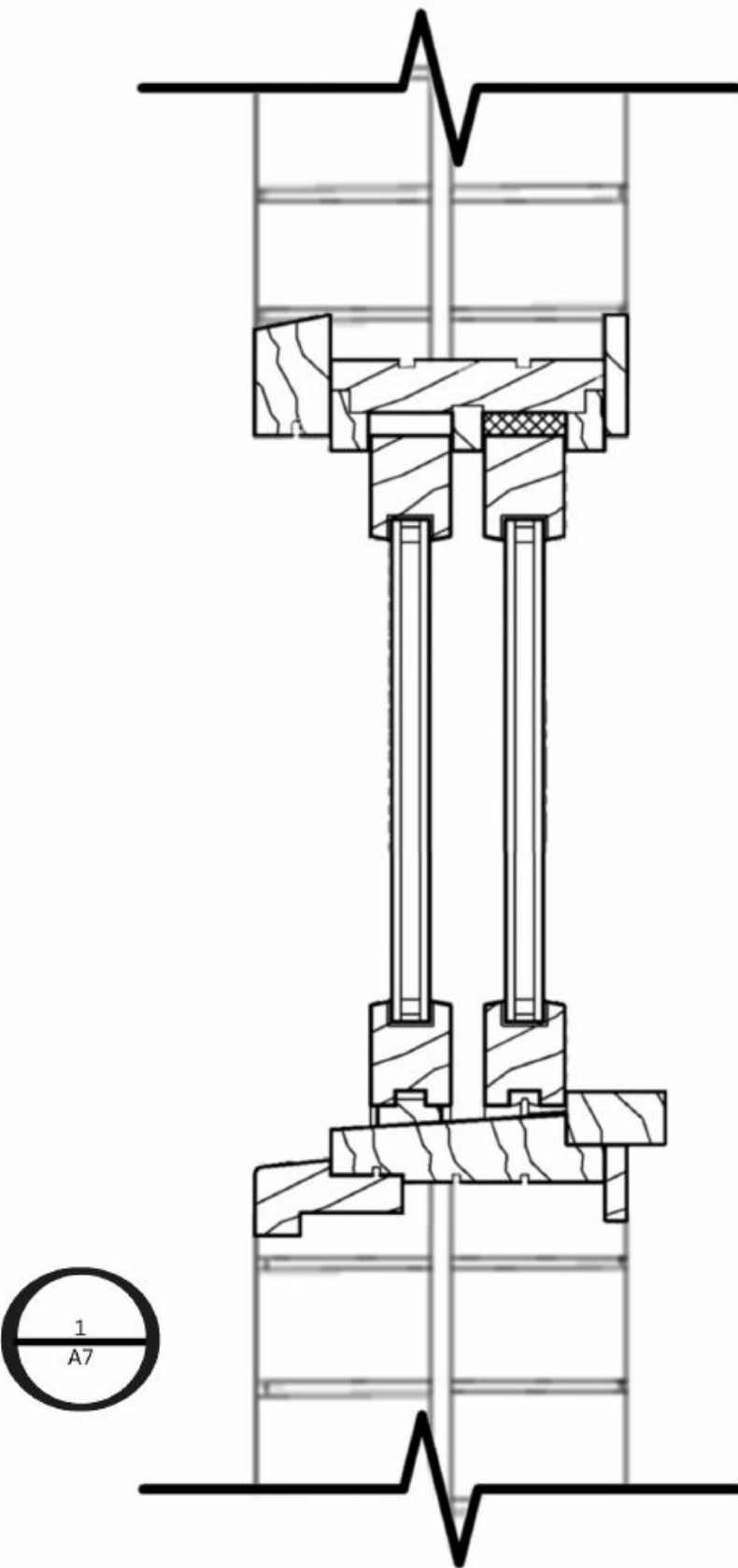


U-VALUE

The improved ecological roof assembly achieves a U-value of 0.205 W/m²K. This is made possible by the inclusion of wood-fibre insulation between and above the rafters, which boost resistive value and reduce thermal bridging. Together, these changes create a roof that is both energy-efficient and aligned with natural, carbon-storing materials.

CARLETON UNIVERSITY AZRIELI SCHOOL OF ARCHITECTURE AND URBANISM	EXPERIMENTAL HOUSE MELALAMMETIE, 40900, FINLAND	ENVELOPE RE(DESIGN) AND IMPACT ASSESSMENT	SHEET NUMBER:	TEAM MEMBER: Nooraiz Noorani	REVISION SCHEDULE:	11/10/2025	Nooraiz Noorani	Initial iteration of Ubekus
	DATE OF CONSTRUCTION 1952-21954		A6			11/14/2025	Nooraiz Noorani	Added Roof Analysis, Diagrams, Drawing, and Graphs
						11/24/2025	Nooraiz Noorani	Modified details, general sheet layout after pin ups and TA feedback
						11/25/2025	Nooraiz Noorani	Modified drawings and added final details

ORIGINAL WINDOW ASSEMBLY



- Window Specifications** - Original 1950s Window
- Two separate single-glazed frames (double-sash)
 - Non-thermally broken wood frame
 - Simple wooden spacer
 - Air filed gap
 - No coating

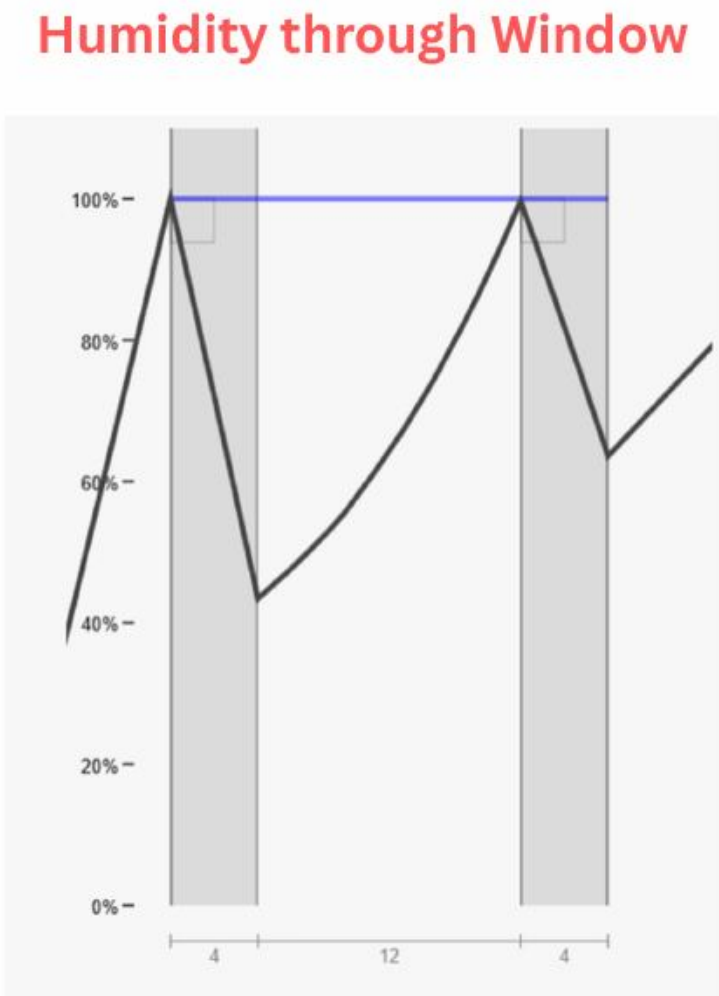
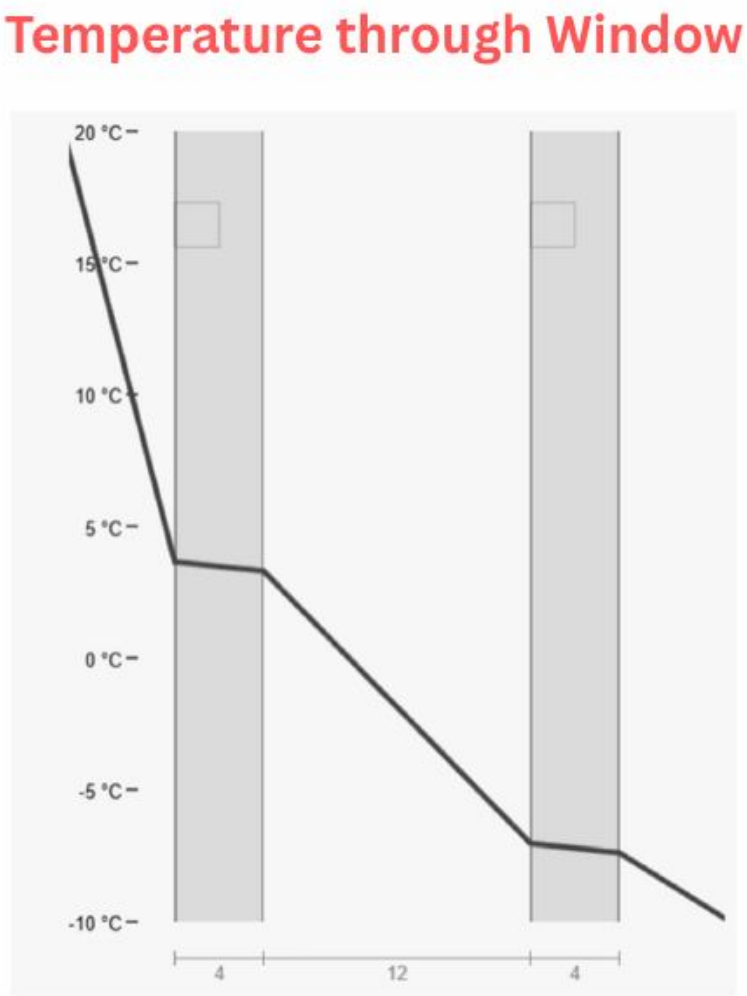
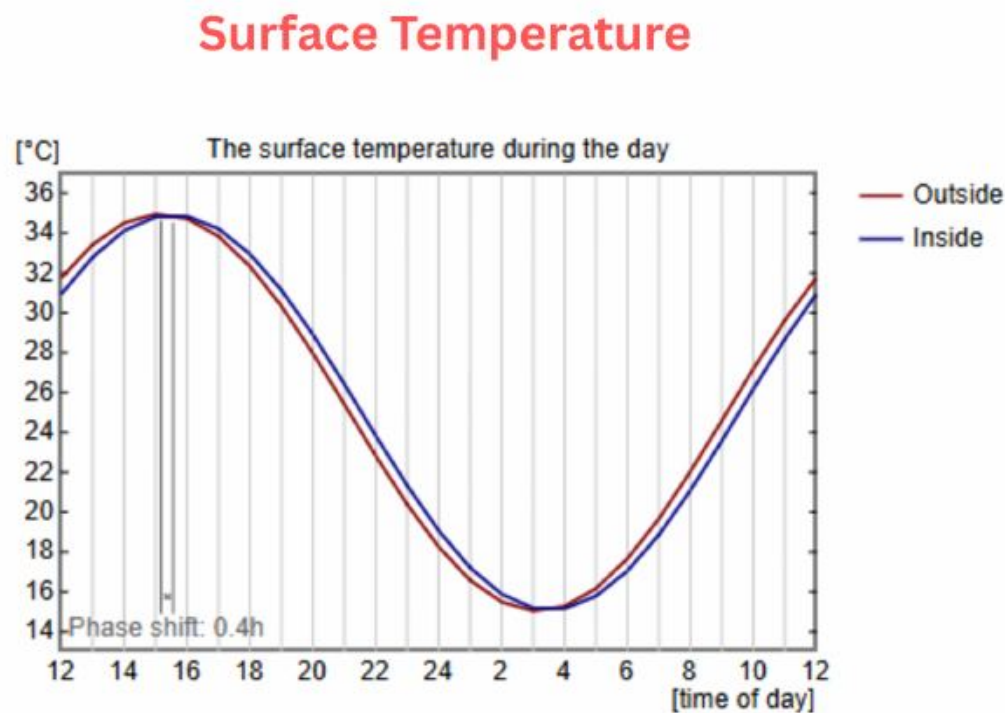
Performance Metrics - LBNL Window

U-Factor: 2.810 W/m²·K

Solar Heat Gain Coefficient (SHGC): 0.5313

Visible Transmittance (VT): 0.561

The original Muuratsalo Experimental House window, modeled as a simple double-glass assembly, demonstrates moderately bad thermal performance. Its U-factor was calculated to be 2.810 W/m²·K using LBNL Window, and 2.954 W/m²·K with Ubakus, reflecting the limited insulation provided by the air-filled cavity and absence of low-E coatings. Solar heat gain and visible transmittance are relatively high, which allows daylight penetration but contributes to heat loss during colder months. Overall, the window provides basic thermal separation but is energy inefficient by modern standards.



U-VALUE

Ubakus

R-VALUE

0.130

0.005

0.158

0.005

0.040

LCA

Traditional 1950s double-sash wood windows have low embodied energy due to their simple timber construction and long repairable lifespan. However, untreated or aging wood is vulnerable to moisture-related decay, warping, and air leakage, which can reduce performance over time and require more frequent maintenance. Although individual parts can be repaired, ongoing deterioration often limits full restoration and results in most end-of-life timber being suitable only for low-grade reuse or disposal.

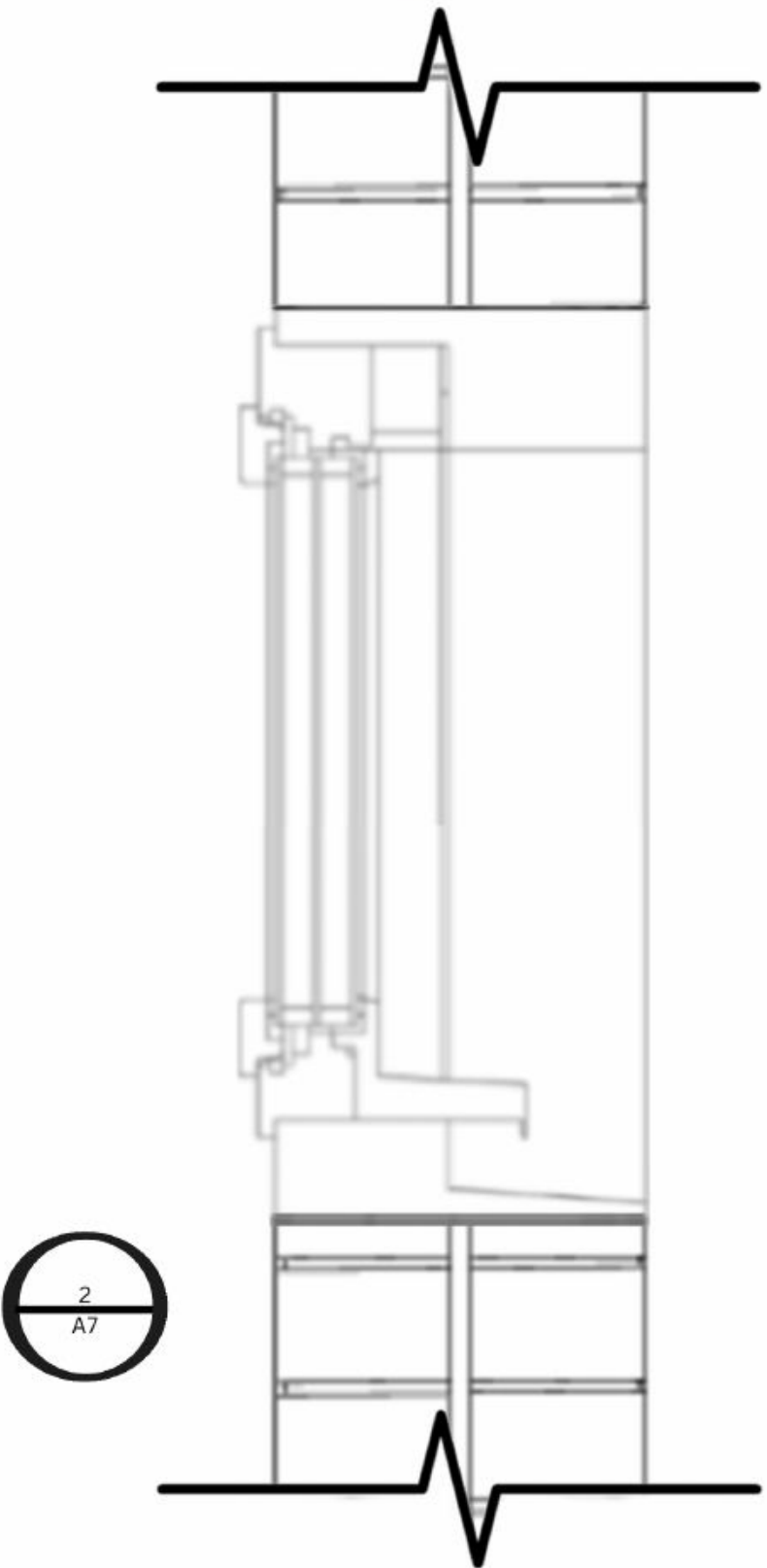
TOTAL R-VALUE

0.339

TOTAL U-VALUE

2.954

THERMAL/CARBON PERFORMANCE WINDOW ASSEMBLY



- Window Specifications** - Modern Efficient Window
- Triple glazed insulated (IGU)
 - Thermally broken aluminum/wood frame
 - Warm-edge thermally improved spacer
 - Argon filed
 - Low-E Coating

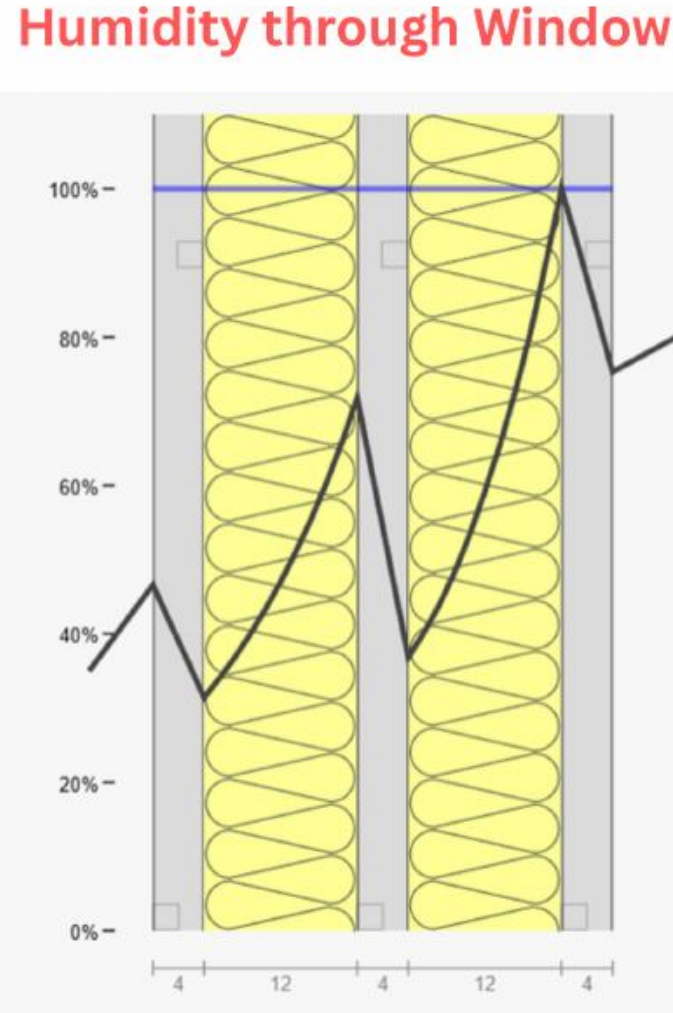
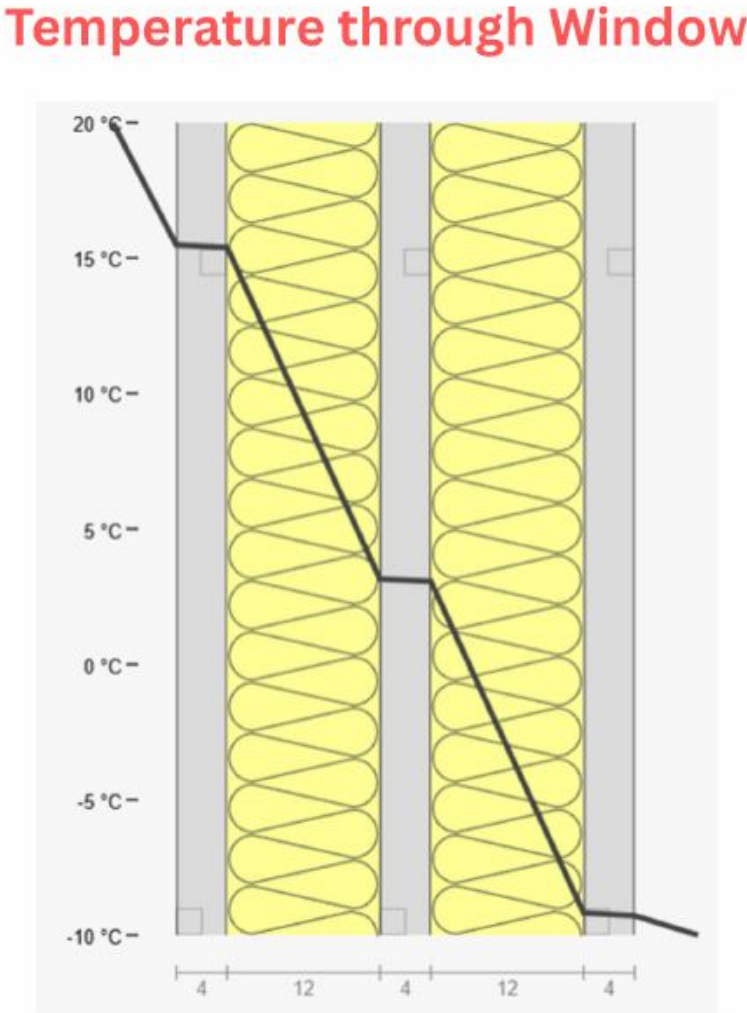
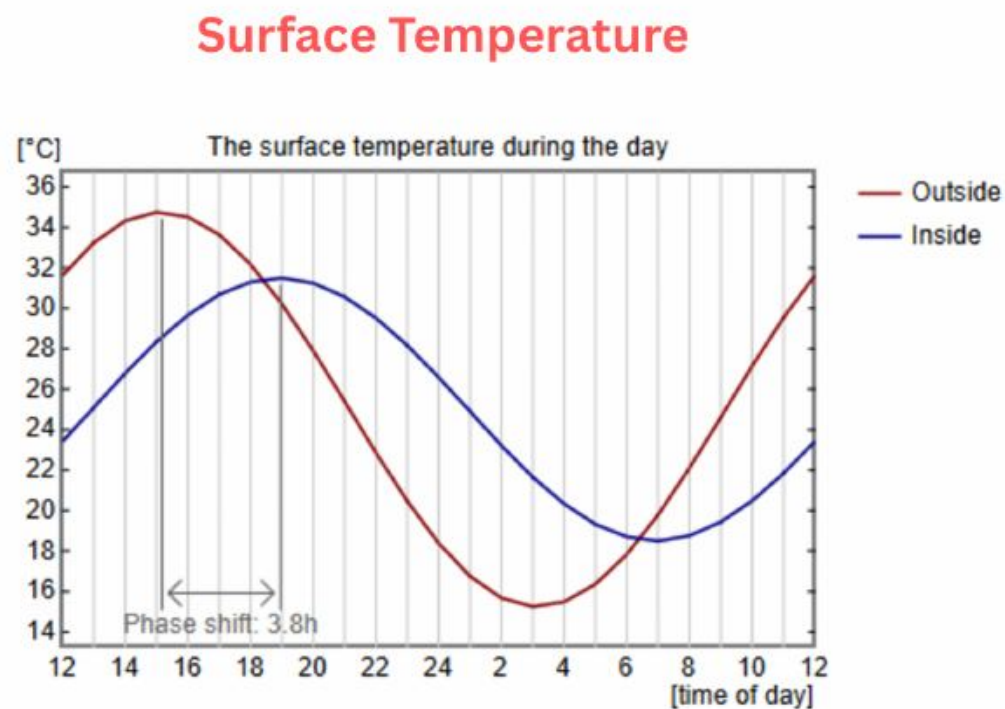
Performance Metrics - LBNL Window

U-Factor: 1.679 W/m²·K

Solar Heat Gain Coefficient (SHGC): 0.290

Visible Transmittance (VT): 0.436

The upgraded triple-glazed window significantly improves thermal performance. This analysis shows the carbon version, with the difference being the sourcing/production/transportation/installation of green materials while using sustainable practices, leading to slightly lower insulation compared to the thermal version. The window’s U-factor was calculated as 1.679 W/m²·K for LBNL Window, and 0.649 W/m²·K for Ubakus, showing the combined benefits of an argon-filled cavity, an additional glazing layer, and low-E coatings. Solar heat gain is reduced, which helps limit overheating in summer, while visible transmittance remains sufficient for daylighting. Compared to the original window, this assembly provides better insulation, reduced condensation risk, and improved overall energy efficiency.



U-VALUE

Ubakus

R-VALUE

0.130

0.005

0.677

0.005

0.677

0.005

0.040

LCA

A modern recycled-aluminum and wood hybrid frame carries higher manufacturing impacts but offers excellent durability and recyclability. Combined with triple glazing, low-E coatings, and argon fill, these windows dramatically reduce heat loss, yielding significant operational energy savings that typically outweigh the higher embodied impacts over the building’s life cycle.

TOTAL R-VALUE

1.54

TOTAL U-VALUE

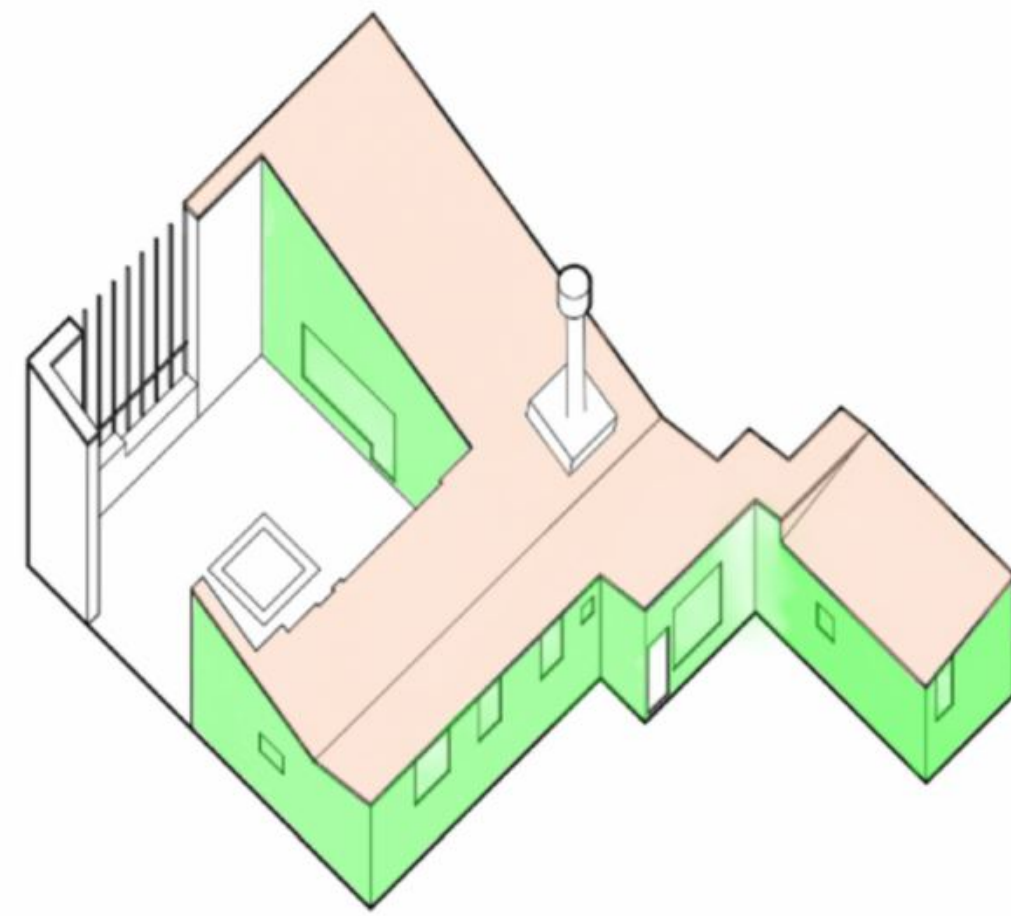
0.649

CARLETON UNIVERSITY AZRIELI SCHOOL OF ARCHITECTURE AND URBANISM	EXPERIMENTAL HOUSE MELALAMMETIE, 40900, FINLAND DATE OF CONSTRUCTION 1952-21954	ENVELOPE RE(DESIGN) AND IMPACT ASSESSMENT	SHEET NUMBER: A7	TEAM MEMBER: Colin Murray	REVISION SCHEDULE:	11/14/2025 Colin Murray Added Window Analysis, Diagrams, Drawing, and Graphs	11/24/2025 Colin Murray Modified details, general sheet layout after pin ups	11/25/2025 Colin Murray Modified drawings, final details

BASED ON 2021 IECC RESIDENTIAL BUILDING CODE TABLE R402.1.2 : MARINE 4

Both the thermal and carbon iterations outperform IECC code requirements. The final decision for the retrofit of this home would include cost analysis of the materials as well as how these changes would impact annual cooling and heating loads. Performing better than code is not the main goal of this retrofit, we must keep in mind the larger impact on both occupant comfort and GHG emissions.

A 3D isometric illustration of a red building with a white antenna on its roof and a white fence on the left. The building has a complex, multi-angled roofline. The antenna is a white, cylindrical structure with a small sphere on top. The fence is a white, vertical-bar structure. The building is shown from a perspective that highlights its red color and the white details.



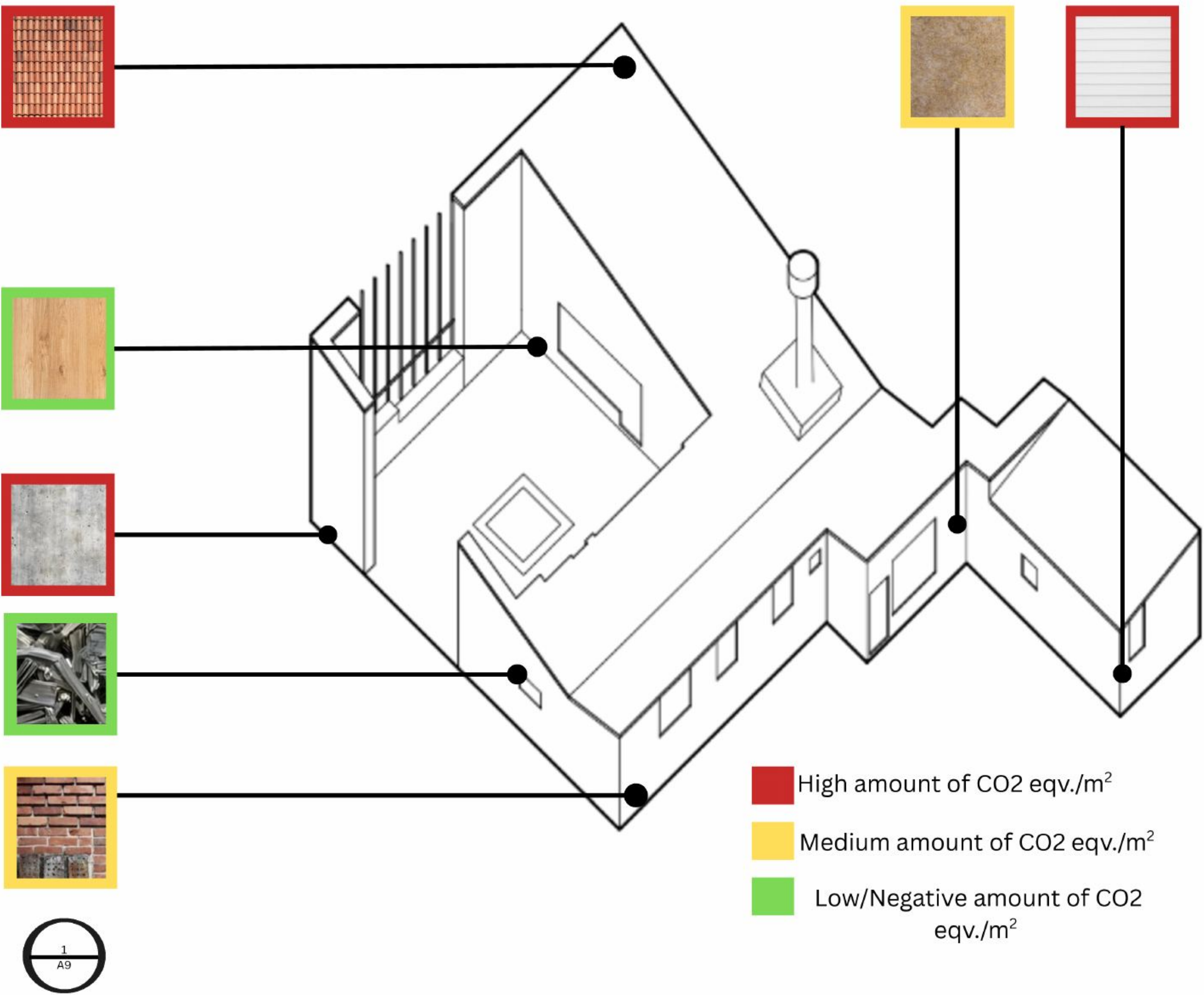
Does not meet IECC code



CARLETON UNIVERSITY AZRIELI SCHOOL OF ARCHITECTURE AND URBANISM	EXPERIMENTAL HOUSE MELALAMMETIE, 40900, FINLAND	ENVELOPE RE(DESIGN) AND IMPACT ASSESSMENT	SHEET NUMBER:	TEAM MEMBER: Manreet Atwal & Kaitlin Gibson	REVISION SCHEDULE:	Kaitlin Gibson	11-20-2025	Formatted table and compiled data
	DATE OF CONSTRUCTION 1952-21954		A8			Manreet Atwal	11-25-2025	Created 2D and 3D Diagrams

LCA OF EXISTING MATERIALS

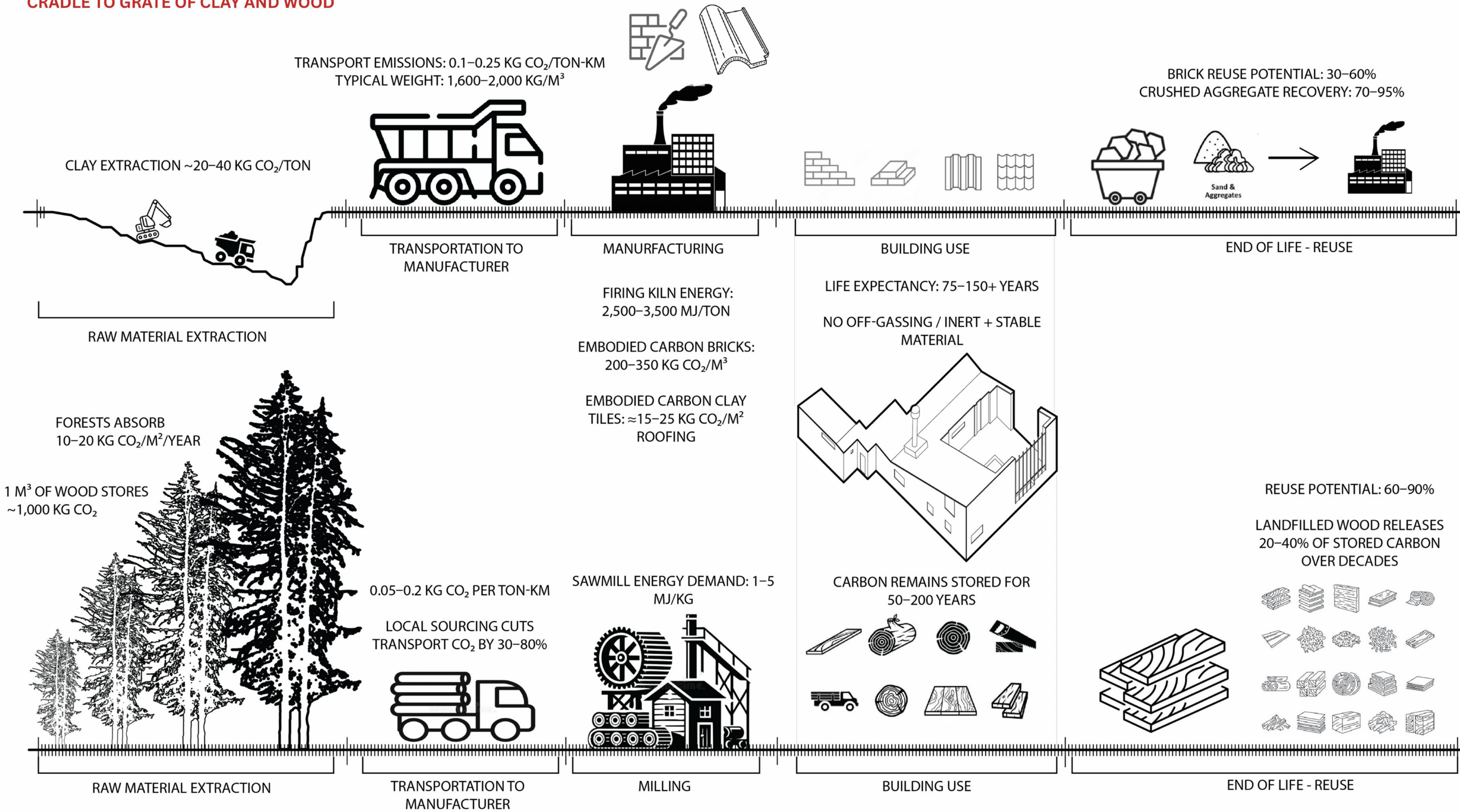
Material		Raw Material and manufacturing	Transportation	Waste from Construction/ Installation	Maintenance/ Replacement	End of Life
Wood		Forestry-milling, - kiln-shaping/sawing - drying	Extraction location → sawmill → site	Deformed wood, leftover waste, irregular pieces	Minimal, replacement only	Strong reuse potential
Concrete		Cement production, aggregate extraction	Quarry/pit → plant → site	Extra from pouring and redi-mix trucks	None	Strong recycle potential, reused as aggregate
Rockwool Stone Wool		Volcanic rocks - slag- kiln	Extraction/mining → plant → site	Cut/shaping waste on site	None	Landfill, no reuse potential
Clay Roofing Tiles		Drying and high heat firing	Extraction location → manufacturing plant → site	Cut/shaping waste	50 Year Life	Recycled as crushed aggregate (grog)
Recycled Aluminum		Collection - treatment - remolding - assembly	Collection → industrial plant → assembly factory → site	Manufacturing dross/chemical wastewater	Minimal to none	Up to 90% reuse
Original Brick		Originally collected form local structures	Collection locations → site → new site location	Leftover mortar waste, cut/shaping waste	Occasional, mortar retouching	Strong reuse potential, crushed for aggregate
Siding		Petroleum extraction - PVC resin production - mixed with additives - extruded into siding.	Lightweight panels → truck or ship → site.	Offcuts of vinyl siding, cracked panels, packaging waste	Durable, low upkeep, no painting, long lifespan.	Hard to recycle, not biodegradable, toxic if burned.



The above diagram represents the effects of the materials used in the original construction of the building and the environment. The impact is quantified in kg of CO2 eqv./m² of material for a cradle to grave analysis.. The images highlighted in red represent images with a high CO2 eqv. in all aspects of the LCA. The images in yellow include the exterior brick and insulation. The brick is highlighted in yellow as the production of brick has a high carbon content but this project reused bricks from another site. The purpose of this analysis is to be able to quickly identify which materials are having the largest impact of GHG emissions.

CARLETON UNIVERSITY AZRIELI SCHOOL OF ARCHITECTURE AND URBANISM	EXPERIMENTAL HOUSE MELALAMMETIE, 40900, FINLAND DATE OF CONSTRUCTION 1952-21954	ENVELOPE RE(DESIGN) AND IMPACT ASSESSMENT	SHEET NUMBER: A9	TEAM MEMBER: Nooraiz Noorani and Kaitlin Gibson	REVISION SCHEDULE:			
--	--	---	-------------------------	--	--------------------	--	--	--

CRADLE TO GRATE OF CLAY AND WOOD



CARLETON UNIVERSITY AZRIELI SCHOOL OF ARCHITECTURE AND URBANISM	EXPERIMENTAL HOUSE MELALAMMETIE, 40900, FINLAND DATE OF CONSTRUCTION 1952-21954	ENVELOPE RE(DESIGN) AND IMPACT ASSESSMENT	SHEET NUMBER: A10	TEAM MEMBER: Nooraiz Noorani	REVISION SCHEDULE:			
--	--	---	--------------------------	-------------------------------------	--------------------	--	--	--